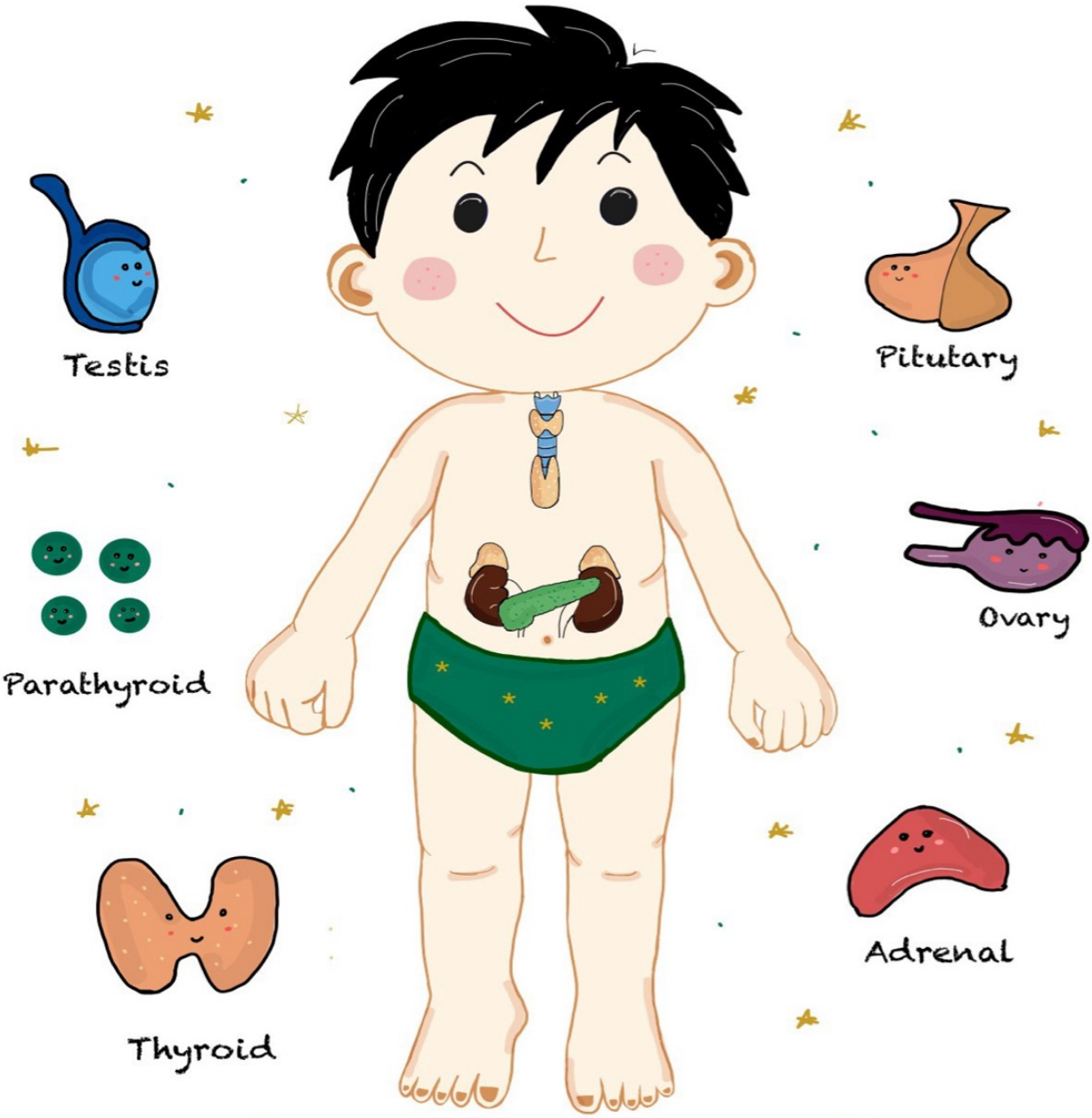
PEDIATRIC ENDOCRINE SYSTEM



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- Pituitary gland disorders (Hypopituitarism)
- Thyroid gland disorders → Congenital hypothyroidism
  → Acquired hypothyroidism
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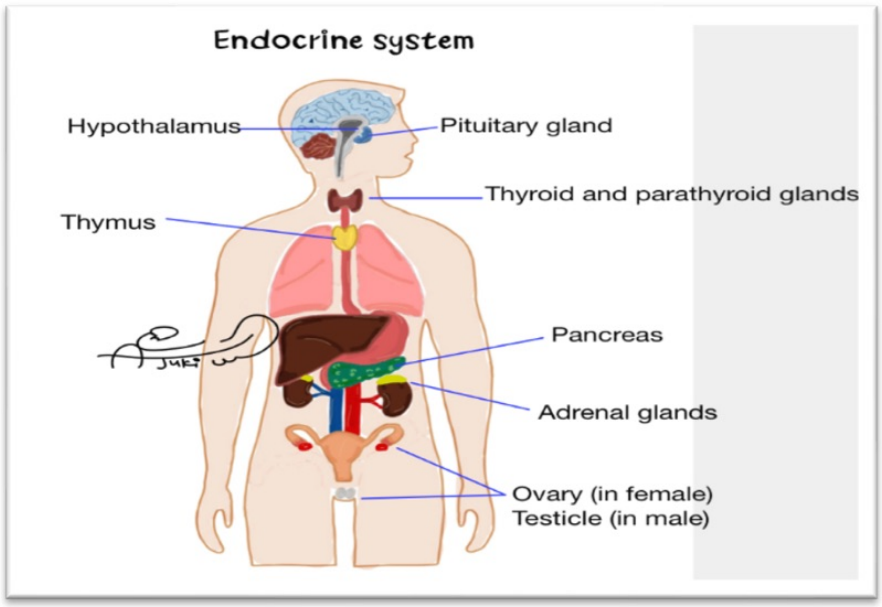
- Adrenal gland disorders → Adrenal insufficiency
  → Cushing syndrome
  → Congenital adrenal hyperplasia
- Diabetes mellitus & Examination Related
- Hypoglycemia in children
- Diabetes insipidus
- Short stature + (History of SS – Examination of SS)
- Puberty → Precocious puberty
  → Delayed puberty
- Parathyroid gland disorders → Hypo para
  → Hyper para

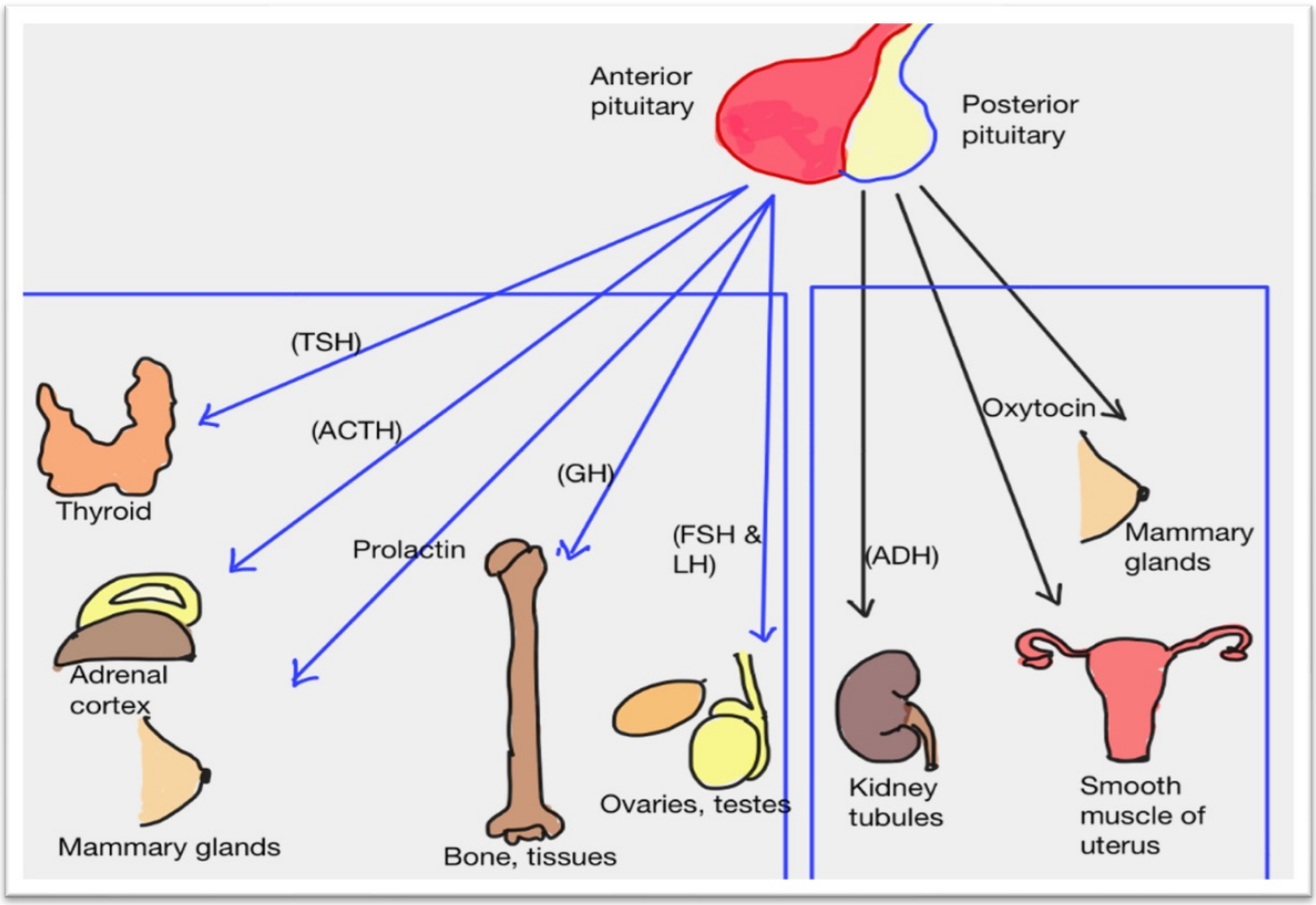
The Main Endocrine glands

1. Pituitary gland
2. Thyroid gland
3. Adrenal gland
4. Parathyroid gland
5. Ovary & Tests
6. Pancreas



Control of Hormone Release:

1. Feedback mechanism
2. Hormonal stimuli
3. Nervous system (Hypothalamus)



HYPOPITUITARISM

Physiology of Growth hormone:

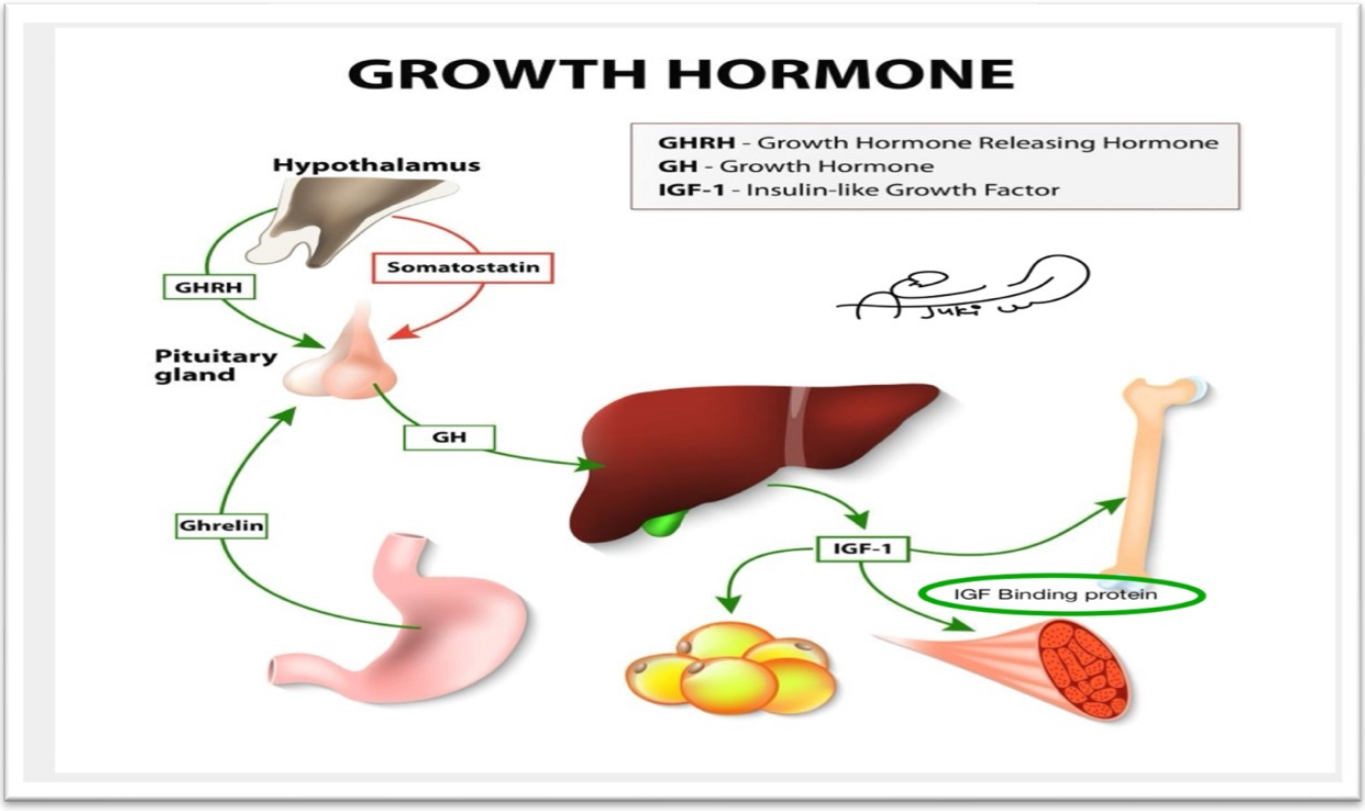
- GH is a polypeptide hormone consisting of **191 amino acids** secreted by somatotroph cells in the Anterior pituitary.
- GH secretion is **pulsatile (so random sample for GH level is of a little valve)**

Factors Increase secretion of GH:	Factors decrease secretion of GH:
1. Stress – Sleep – Exercise 2. Hypoglycemia – starvation 3. α adrenergic stimulation 4. Arginine	1. GH therapy 2. Hyperglycemia 3. B Adrenergic stimulation 4. Somatostatin

Action of GH:

1. Increase linear growth (Increase bone growth) – soft tissue growth, Ms.
2. Stimulate protein synthesis
3. Anti-insulin Effect + (Hyperglycemia, + lipolysis

NB: (Action is mediated by insulin like growth factor-1 (IGF-1) formed by liver

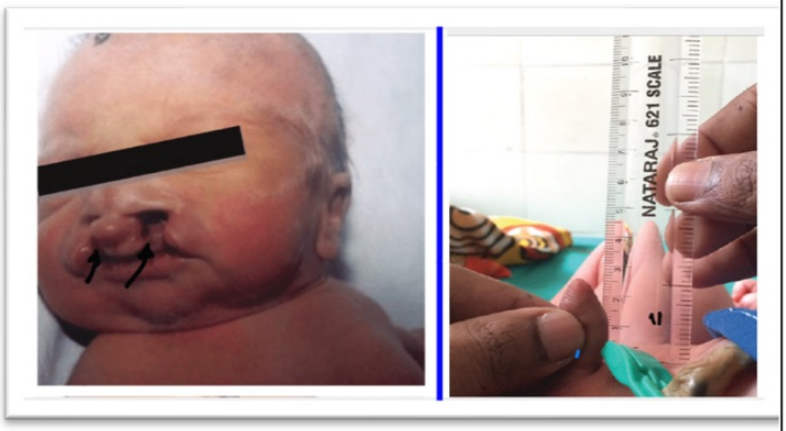


Hypopituitarism– denotes:

1. Underproduction of GH alone.
2. In combination with deficiencies of other pituitary hormones.

- Incidence 1 in 4000 children (Pituitary disorders are **uncommon** in childhood).
- NB (Decrease of all anterior pituitary hormones is called panhypopituitarism).
- Onset → at any time from neonatal period – childhood
- Hypopituitarism generally means (growth hormone deficiency GHD)

Causes of hypopituitarism:

Isolated GH deficiency.	Panhypopituitarism.
Genetics: 1. Mutation of GH or GHRH genes 2. End Organ resistance: • Abnormal GH receptors • **Laron syndrome** متلازمة لارون (rare condition due to defective GH receptors resulting in GH insensitivity. Patients with this condition have high GH levels but low levels of IGF–1] • Abnormal (IGF 1) **Acquired:** 1. Idiopathic 2. Post cranial irradiation	**Genetics:** 1. Mutation of multiple pituitary hormone genes **Congenital:** 2. Pituitary gland agenesis or Hypoplasia (usually ass with midline defect)  **Acquired:** 1. Tumor (**craniopharyngioma)** 2. Trauma–post cranial surgery 3. Post infection (post meningitis) 4. Cranial irradiation 5. Infiltration, e.g. histiocytosis Structural 6. cerebral malformation, or hydrocephalus

Clinical picture of CHD:

- **At birth & Neonatal period:**

1. **Normal** length & weight **at birth** (B/C GH is not essential for intrauterine growth)
2. Hypoglycemia (severe & persistent) – Apnea – Cyanosis.
3. Microphallus (Diagnostic clue in ♂) (<2cm stretched in term infant)
4. Prolonged neonatal jaundice.
5. Mid facial anomalies as (cleft palate or Septo – optic dysplasia) →pituitary gland aplasia??

- **Older infant & children:**

1. **Short Stature:**
   - Growth failure with delayed skeletal maturation
   - Appear at the end of 2nd year of life
2. **Childish** Face:
   Immature appearance
   - Small face, Nose, Mandible
   - Wide AF – Fine hair
   - Delayed teething
3. **Hypogonadism & Delayed Puberty**
4. **Hypoglycemia**
5. Weight gain with relative **truncal obesity, increased SC fate.**

Child with GHD has normal mentality and a normal IQ. (mcq)

- **C/P** of other hormonal decrease If panhypopituitarism (Low T_3 T_4 – Low cortisone)

Investigation:

1. GH Level (**GH stimulation provocation test**)
   Measures of GH after stimulation by using GH secretagogues as (**Insulin* –Glucagon – Clonidine –Arginine –Exercise**)
2. **IGF I (Low)** – IGF binding protein (Low)

Affected by: A. Liver disease
B. Malnutrition

3. Level of other pituitary hormones (TFT – cortisone level)
4. GHRH stimulation test (To differentiate bet pituitary for hypothalamic causes)
5. **MRI pituitary gland** – in case of hypoplasia or aplasia
6. **Bone age** (Lt hand & wrist X ray → **Delayed** bone age)
7. Detailed eye and fungus examination if you suspect **brain** tumor !!!

Treatment:

- ✓ **Recombinant GH therapy**, which is given by **daily subcutaneous injection** in the **evening (7 days /week)**, till **closure of the epiphysis**.
- ✓ The best response is seen in children with the **most severe GHD**

NB: If the pt have panhypopituitarism other Hormone replacement therapy should be started also.

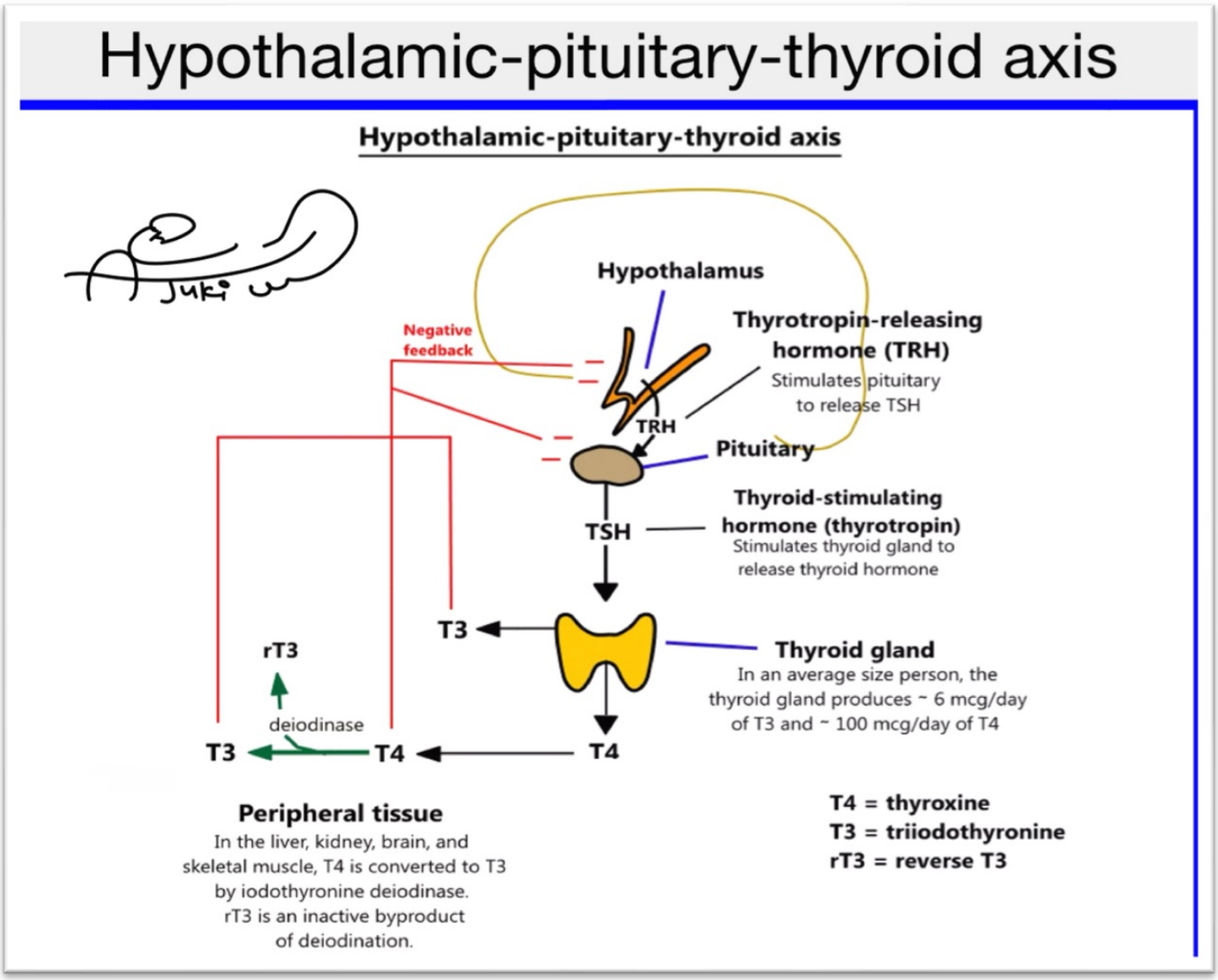
Other indication of GH therapy حفظ	S/E of GH therapy
1. **Turner syndrome (45XO)** 2. **Chronic kidney disease (CKD)** 3. **Prader Willi syndrome** 4. **Idiopathic SS** 5. **Child with IUGR with failure of catch-up growth.** 6. **SHOX deficiency** (±Noonan syndrome, achondroplasia controversial)	1. Pseudotumor cerebri 2. Slipped upper femoral Epiphysis 3. Gynecomastia 4. Increased Insulin resistance 5. Increase risk of Leukemia

- **Recently, recombinant IGF-1** has been used to treat children with GH resistance **(e.g. Laron syndrome)** and IGF-1 deficiency who would have previously not responded to GH treatment.
- Recombinant IGF-1 therapy is **very expensive** and confined to a few specialized centers.

THYROID GLAND

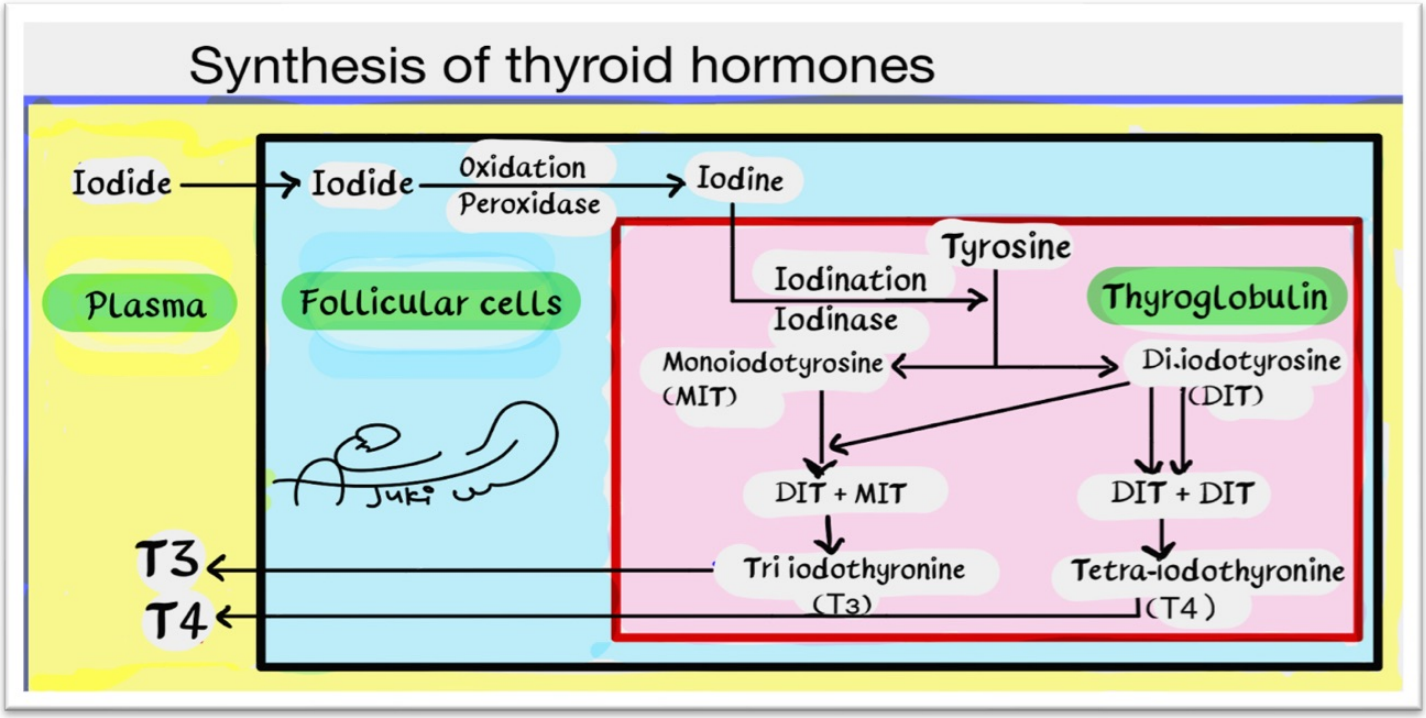
Function of thyroid hormones: " B "

1. Maturation of the **B**rain in the 1st years of life
2. Normal **B**one (skeletal) growth
3. Increase **BMR** (increase glycogenolysis – increase gluconeogenesis increase lipolysis)
4. Inotropic & chronotropic effect on heart (**B**eta adrenergic)
5. Stimulate **g**ut motility, **P**lay role in thermoregulation.

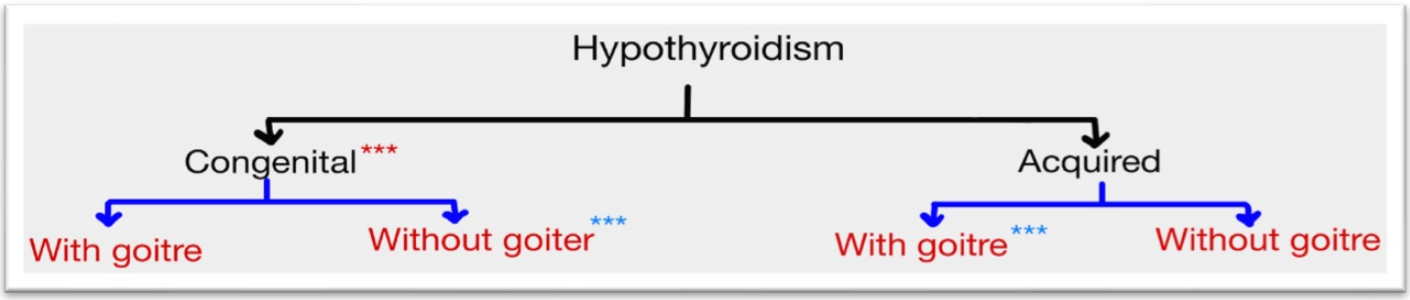


Note:

TSH does not cross placenta, while T_3 & T_4 Cross it in limited amounts, Antithyroid Drugs it can Cross placenta.



HYPOTHYROIDISM



CONGENITAL HYPOTHYROIDISM (cretinism)!!!

Epidemiology:

Incidence: **1: 3000– 4000** live births

Sex: more in ♀ (2:1) , more common in **Down** syndrome.

It's the most common **preventable** cause of **mental retardation** (MR) in children.

Causes:

1. **Primary Hypothyroidism:**

    1. **Thyroid dysgenesis**
        - **Commonest** cause ≃ **90%**
        - Either (**Ectopic** thyroid –Thyroid agenesis – or Thyroid hypoplasia)
        - No goiter

2. Thyroid dyshormonogenesis

- 2nd most common cause 10%.
- AR
- Associated with goiter

Pendred syndrome is an uncommon genetic disease in children.
Triad (Goiter –SN hearing **loss** –**AR** mutation in pendrin protein gene)

3. Maternal thyroid disease (severe maternal hypothyroidism)

- **Transient neonatal hypothyroidism** (disappear within 6m –1y)
- Due to transplacental passage of **maternal TSH blocking Ab** or Anti thyroid drugs

4. Maternal iodine deficiency: a common cause of congenital hypothyroidism worldwide, but rare in the UK/ it can be prevented by **iodination of salt in the diet**

2. Secondary hypothyroidism:

Due to pituitary gland disorder → decrease TSH is rare (<1% of cases)

3. Tertiary hypothyroidism:

Due to hypothalamic disorder → decrease TRH → decrease TSH

Clinical Features:

1. **At birth & neonatal period:**
   - Most infants are **asymptomatic at birth** (Appear Normal) it is difficult & **not reliable clinically** to differentiate from normal in the first month of life but become more prominent with age.
     (**Newborn screening** (Gutheric test) detect & identifies the majority of newborn with congenital hypothyroidism)

Clinical Features: become evident by the age (wks to months after birth):

1. **Coarse** Features: Macroglossia – Large AF – wide PF, noisy breathing – Hoarse cry, Pale, cold, mottled dry skin
2. **Cry little, sleep much**
3. **Prolonged** physiological **jaundice**
4. **Hypotonia – Umbilical hernia – constipation**
5. **Apnea – choking** episodes
6. Goiter (occasionally)as in Dyshormonogenesis –Endemic goiter– Maternal Goitrogenic drugs

C/P in infant & Children:

1. **Delayed growth (short stature)**
2. **Delayed Mental & motor milestones (MR–developmental delay)**
3. **Head** (Delayed Closure AF– Delayed teething)
   (Hair →Coarsely brittle)
   (Macroglossia –Broad depressed nasal bridge)
4. **Skin** → Dry – Pale – cold
5. **Hands** → Decrease HR (bradycardia)
6. **Genitalia** → Delayed sexual maturation. But Precocious Puberty may occur (rare)

Investigation:

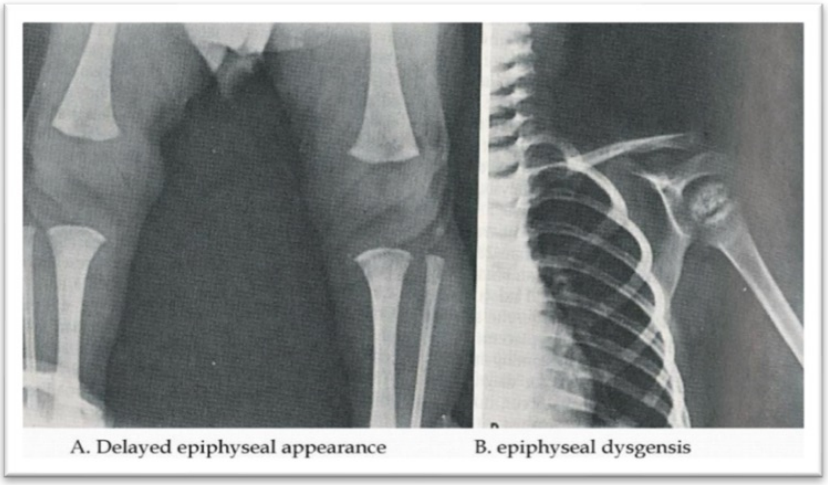
1. **TFT (T_3 –F T_4 – TSH) (Hypothyroidism)**

TFT	Primary (1ry)	Secondary (2ry)	Tertiary (3ry)
T_3 Free T_4 TSH	Low Low **High**	Low Low **Low**	Low Low **Low**
		TRH stimulation test →	to differentiate 2ry from 3ry

2. Thyroid U/S:

   Thyroid Scan:
   - Ectopic → Localized
   - Aplasia → decrease uptake
   - Dyshormogenesis → increase

3. **Knee X Ray** (**Absence** of the distal **femoral epiphysis** in term infant)

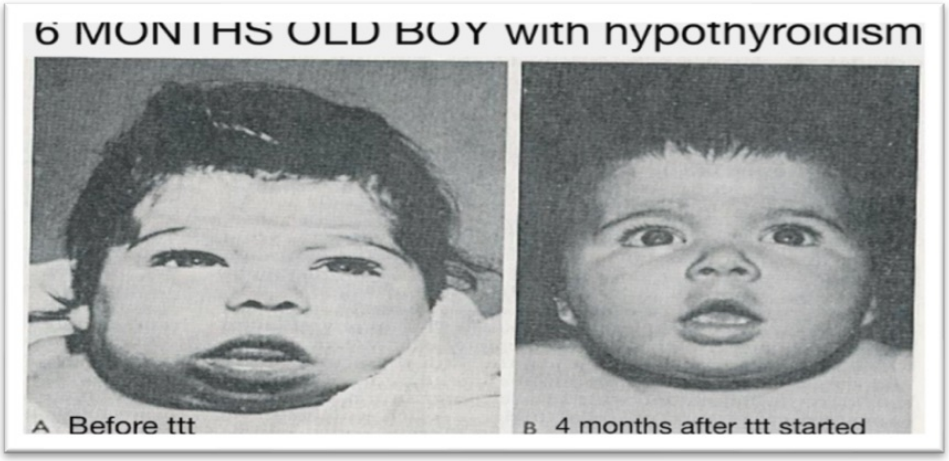

A. Delayed epiphyseal appearance B. epiphyseal dysgensis

4. **Newborn screening**. (Done routinely for all newborn in UK & developed countries for screening (TSH) of congenital hypothyroidism & other genetic disorders.

Treatment:

(Early D_X – Early R_X → good prognosis)

- Levothyroxine (**DOC**) → orally 10 – 15 mcg/kg/Day – **(Life Long)** مدي الحياة



- **Follow-up** (clinically – & Laboratory) in endocrine opd:
  1. **growth**
  2. **TFT**
  3. **Lt Wrist X Ray**
  4. **Development**
  5. **Activity, symptoms**

Treatment started soon after birth (1st 2 wks) reduces the risk of impaired intellectual development.

ACQUIRED HYPOTHYROIDISM

- **Acquired Hypothyroidism** is more common than **hyper**thyroidism in children.
- ≃ 99% of acquired Hypothyroidism is (**Primary**)
- **Hashimoto's thyroiditis** is the most common cause

Other causes of Acquired Hypothyroidism

1. Subacute thyroiditis
2. Iatrogenic (Post thyroid Surgery) – radioactive iodine ttt.
3. Infiltration By tumour
4. Iodine deficiency

Hashimoto's thyroiditis: (Autoimmune thyroiditis)

Is the **most common** cause of sporadic goiter in childhood & adolescence

Epidemiology:

- Sex: more in ♀
- Age: >6 y (10 – 18 y)
- **Positive F/H** of thyroid disease in 25 – 30 % of cases

Etiology:

Autoimmune mediated destruction of thyroid gland cells by:

- **Thyroid autoantibodies:** anti-TPO (anti-thyroid peroxidase) and anti-Tg (anti-thyroglobulin) antibodies, (10-15% of patients with Hashimoto may be antibody negative)

- Associated with other disease
    1. **Autoimmune**: Type I DM – Addison disease –vitiligo –alopecia–celiac.
    2. **Syndromes**: Down Syndrome – Turner Syndrome

C/P: Typically has an **insidious** onset:

1. **Symptoms**:
    - Early **nonspecific** symptoms may include the following (Fatigue–Constipation–Dry skin– Weight gain)
    - Sleep disturbances– cold intolerance
    - Asymptomatic **goitre*** (smooth – non-tender –regular surface –firm)

- Delayed puberty, amenorrhoea, precocious puberty
- Poor concentration–Deterioration in school work– Learning difficulties.

Note (≃ 5–10% may present with C/P of hyper T_3 T_4)

2. Signs:
   - Reduced growth rate/short stature
   - Unexpected weight gain/obesity – Goitre
   - Thin, dry hair– Pale, puffy eyes with loss of eyebrows
   - Dry skin, Cold peripheries– Bradycardia
   - Slow–relaxing reflexes – Slipped upper femoral epiphysis

Investigation:

1. **TFT**: (free T4 **low** –TSH **high**) (may be normal in euthyroid state)
2. Serum Ab (Positive in 85%of cases)
3. Thyroid U/S & isotope Scan (not necessary for diagnosis)

Rx:

- **Levothyroxine sodium** is given in case of **hypo T_3 T_4 for Life**.
- Follow up in endocrine OPD is very important.

The **prognosis** for autoimmune thyroiditis is very **good** and the outlook partly depends on whether the child will develop **other** autoimmune diseases)

HYPERTHYROIDISM

- Thyrotoxicosis is **uncommon** in childhood
- Hyperthyroidism usually results from **'Graves' disease (95%)**

Other causes of hyperthyroidism:

1. Transient neonatal thyrotoxicosis
2. Toxic nodular goitre
3. Mc Cune Albright syndrome

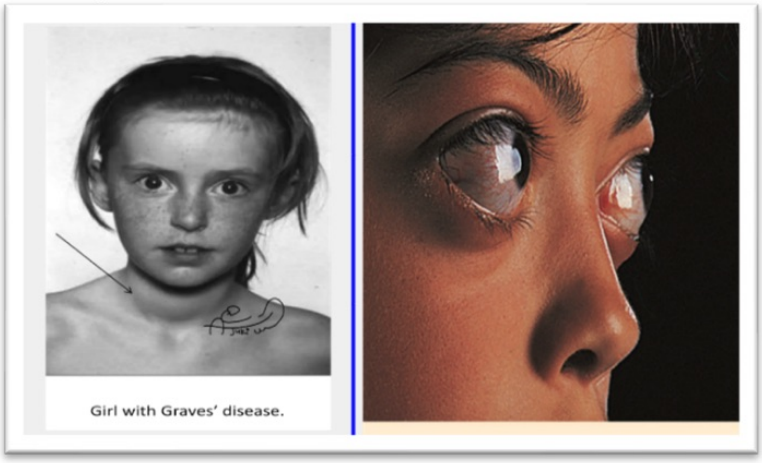
Graves' disease: (Autoimmune thyroiditis)

Cause: Caused by **stimulatory antibodies** to the TSH Receptors

Sex: More **common in female ♀** (teenage female)

Clinical picture:

GD is **multi systemic autoimmune** disorder of involving **(skin – eyes– thyroid gland)**

Systemic	Eye signs (uncommon in children)
• Anxiety, restlessness Sweating • Diarrhea & Weight loss • Rapid growth in height(tall) • Advanced bone maturity • Tremor & Tachycardia, wide pulse pressure • Warm, vasodilated peripheries • Goitre (with bruit) • Reduced concentration, • Learning difficulties / behavior problems	*Exophthalmos & Ophthalmoplegia *Lid retraction *Lid lag  Girl with Graves' disease.

Investigation:

1. **TFT** (free T_4 **high**, TSH **low**)

2. Thyroid **antibodies**:
   - Thyroid peroxidase Ab
   - TSH receptors Ab (+ve 95% of cases)

3. Imaging Thyroid ultrasound scan & radionuclide scans (high uptake)

Treatment:

1. Medical (1st Line).

(1)– Carbimazole or propylthiouracil

1–There is a risk of **neutropenia** from **antithyroid medication**, and **all families** should be **warned** to seek **urgent** help and a **blood count** if **sore** throat and high **fever** occur **while** on **treatment**.

2–**medical** treatment is given for about 2 years, which should control the thyroid hormone excess, but **eye signs** may not resolve (**relapse occurs in ⅔ 70% of patients**)

3–Follow–up is always required as thyroxine replacement is needed for **subsequent hypothyroidism** after 1st or 2nd line therapy.

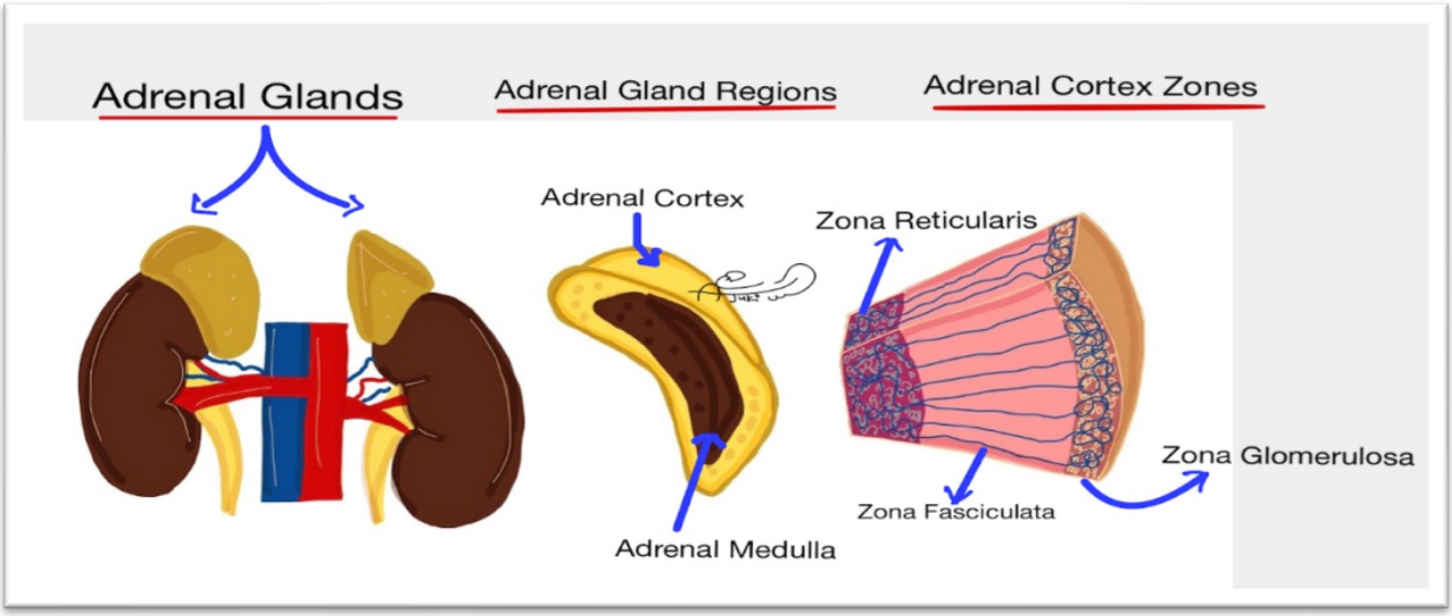
(2)–Propranolol: started **initially** to give **symptomatic** relief of anxiety, tremor, and tachycardia.

2. **radioiodine treatment** (older child) **or surgery** (subtotal thyroidectomy) (**2nd Line**)

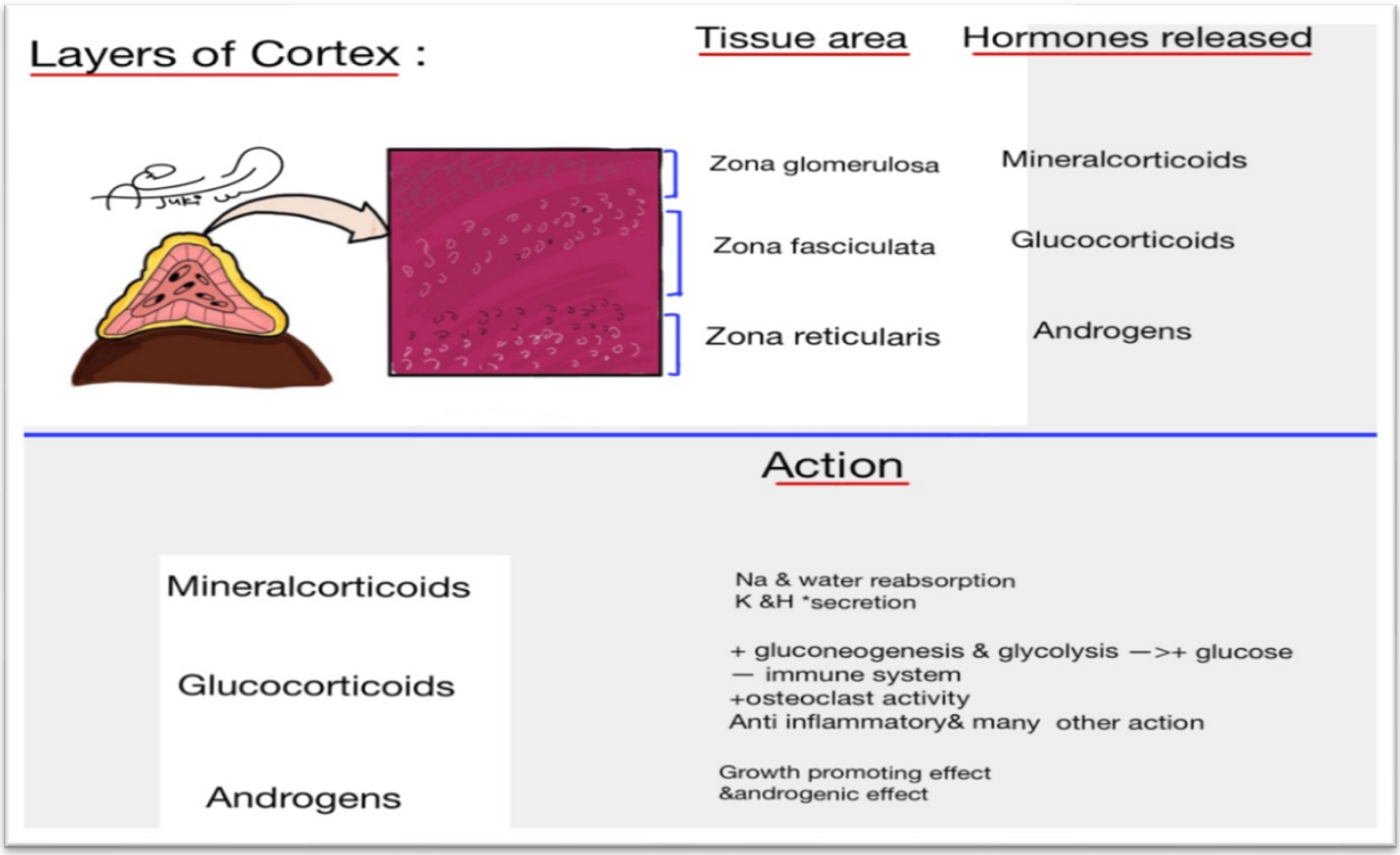
Transient Neonatal hyperthyroidism (may occur in infants of mothers with hyperthyroidism from the transplacental transfer of thyroid–stimulating hormone receptor antibodies. Treatment of the neonate is required as it is potentially fatal, but it resolves with the regression of maternal antibodies).

ADRENAL GLAND

Anatomy & Physiology:



- The glucocorticoids & androgen is under the control of ACTH via the hypothalamic – pituitary – Adrenal Axis
- The Aldosterone is influenced by salt & water balance via the Renin – angiotensin axis



CUSHING SYNDROME

- The term (**Cushing's syndrome**) is used to describe a condition resulting from **long-term exposure to excessive glucocorticoids.**

Etiology: either (exogenous or endogenous)

1. Exogenous is the most common cause of administration of **exogenous** corticosteroids (**Iatrogenic**) as in nephrotic syndrome– Br asthma.

2. Endogenous: rare in children.

A. ACTH independent (1ry): Adrenal tumor, Adenoma. Carcinoma, Adrenal Hyperplasia
B. ACTH dependent (2ry): **Pituitary** gland adenoma (**Cushing disease**)
Ectopic ACTH secreting tumor

Clinical picture:

1. Increase weight (Obesity)– Central and truncal
   - Moon face – buffalo hump
2. Growth failure → short stature + decrease growth velocity– Delayed puberty
3. Virilization – hirsutism – striae – acne.
4. polycythemia (Red cheeks)– easy bruising.
5. Hypertension–Insulin resistance.
6. Ms Wasting – osteoporosis–Osteopenia – psychological problems.

Diagnosis:

(The diagnosis of Cushing syndrome is often questioned in obese children + short stature)

1. 24 hours urine free cortisol – high (triple)
2. Serum cortisol level – loss of normal diurnal variation (double increase midnight value)
3. Dexamethasone suppression test (administration of dexamethasone at night–time → failure to suppress the morning 9,00 hr plasma cortisol level)
4. Plasma ACTH level (1ry or 2ry)
5. RBS high – Low K – Metabolic Alkalosis.
6. CT scan Abdomen or MRI pituitary → if u suspect Tumor

Rx:

1. Cushing syndrome is treated by gradual withdrawal of exogenous steroids.
2. If endogenous the management depends on the source but is usually surgical resection and/or radiotherapy.

ADRENAL INSUFFICIENCY

Causes:

Primary	secondary
Acquired: 1. **Addison's disease** 2. Adrenal hemorrhage /infarction (Waterhouse–Friedrichsen disease) – neonatal, meningococcal, septicaemia 3. tuberculosis TB 4. Post adrenal surgery **Congenital:** 1. **Congenital adrenal hyperplasia** 2. XLR adrenoleukodystrophy a rare neurodegenerative metabolic disorder **Idiopathic**	**pituitary**: A. Panhypopituitarism B. Post– cranial Surgery – Trauma – Tumor C. Long term Glucocorticoid therapy

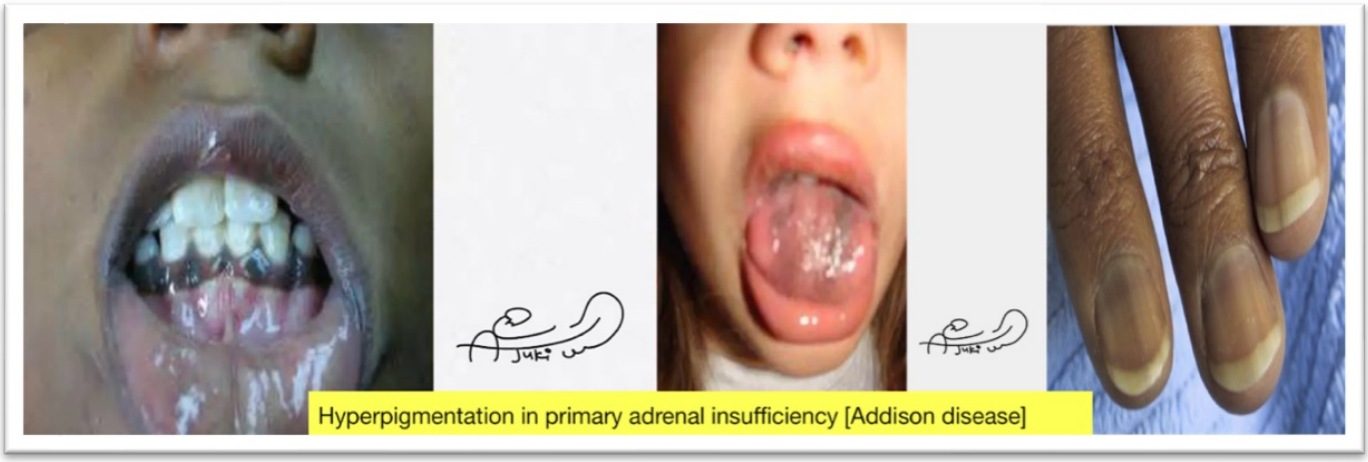
ADDISON'S DISEASE

- It's 1ry adrenal insufficiency (**Autoimmune disorder**)
- Associated with other autoimmune disease as (**Type 1 DM – celiac Hashimeto's – vitiligo**)

C/P:

Usually **insidious (non-specific symptoms)** but (acute adrenal crisis may occur)

1. Fatigue – Weakness– Dizziness – Weight loss
2. Nausea – Vomiting – Syncope
3. Postural Hypotension
4. Skin – Hyperpigmentation in the palm creases (increase ACTH) – mucus membrane of the mouth, skin scars.



Hyperpigmentation in primary adrenal insufficiency [Addison disease]

(It may present more **acutely** with hypovolemic shock and hypoglycemia during intercurrent illness)

→ **adrenal crisis**

Diagnosis:

1. Electrolytes (Na low – K high)
2. Blood sugar (low)
3. ABG (Metabolic acidosis)
4. Serum cortisol level (Low), ACTH level (High) except in pituitary dysfunction.
5. **ACTH stimulation test (Synacthen test)** (cortisol concentrations remain low in primary adrenal dysfunction. A normal response excludes adrenal insufficiency)

Treatment:

Long term Hormonal replacement therapy for life (Hydrocortisone & Fludrocortisone).

NB: (stress dose) 2*or 3* dose of hydrocortisone during febrile illness, gastroenteritis, major trauma, surgery to prevent adrenal crisis.

Adrenal crisis (EMERGENCY)

C/P:

1. Nausea – Vomiting + diarrhea (Low BP)
2. Dehydration – Vascular collapse & shock

Investigation:

1. U/E (Na – Low <130 – K high >5)
2. RBS (Low)
3. ABG (Metabolic acidosis): PH → Low
   Hco3 → Low

Rx:

- ✓ ABC
- ✓ **3 S:**

1–Saline –20 ml /Kg (Normal saline 0,9%) bollus.
2–Sugar – 2–5 ml/ Kg (D10%) IV
3–Steroid – Hydrocortisone IV

"AMBIGUOUS GENITALIA"–Disorders of sex development

Definition:

When the external genitalia do not have the typical anatomic appearance of normal male or female genitalia. = (if you can't determine newborn sex by inspection)

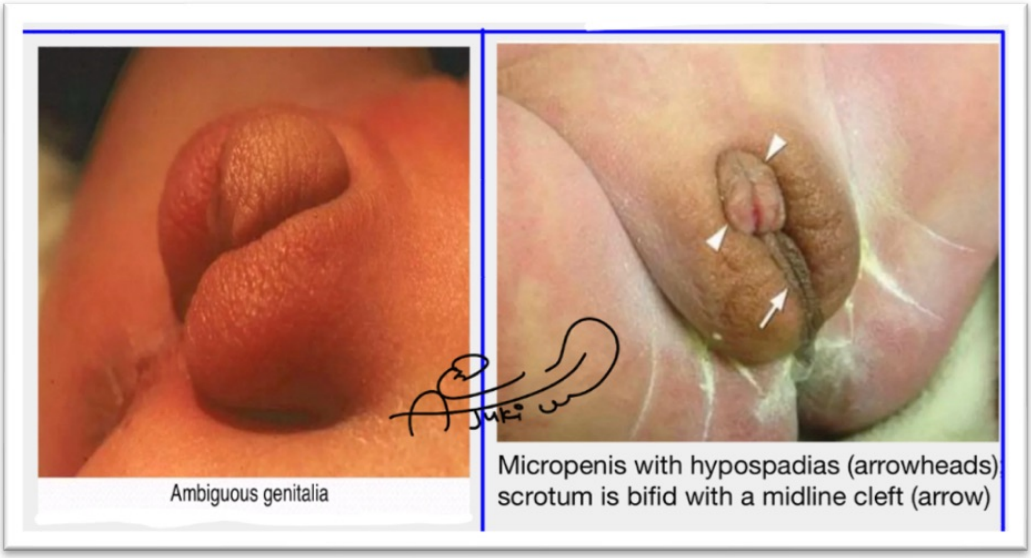
(It is a social & medical emergency)

The fetal gonad is initially bipotential [In the male, a testis–determining gene on the Y chromosome (SRY) is responsible for the differentiation of the gonad into a testis. The production of testosterone and its metabolite, dihydrotestosterone, results in the virilization of male genitalia. & In the absence of SRY, the gonads become ovaries and the genitalia female.

When to Suspect?

1. ♀ external gentalia with clitoromegaly (>1 cm) with inguinal hernia.
2. Micropenis (<2,5cm)
3. cryptorchidism & severe hypospadias

4. Asymmetry of labio scrotal folds–or overtly abnormal genitals like cloacal exstrophy.
5. Perineal hypospadias & bifid scrotum



Causes:

1. **Congenital Adrenal Hyperplasia** (excessive androgens producing virilization in a 46XX female)
2. Androgen insensitivity syndrome (inadequate androgen action, producing undervirilization in a 46 XY male)
3. 5 α reductase deficiency
4. Drugs antenatal.

Approach to a case with Ambiguous genitalia (parents counseling):

1. Good History (**Don't Give name immediately to the baby**)
2. Examination of external genitalia
3. U/S Abdomen of the internal structures and gonads
4. **Karyotyping:** 46 XX –♀
   46 XY – ♂
5. Investigations of the underlying cause (**17HO progesterone**)
6. multidisciplinary team management & psychological support.

If there is abnormal sexual differentiation at birth:

1. **do not guess the infant's sex**
2. the **most common** cause of a disorder of sex development is female virilization from **congenital adrenal hyperplasia**

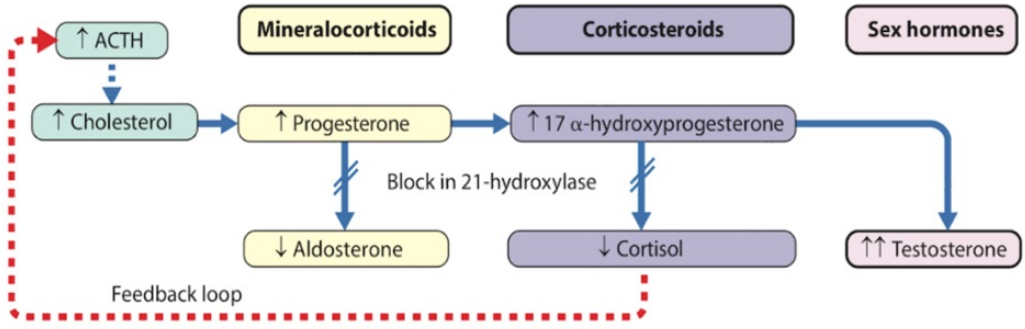
CONGENITAL ADRENAL HYPERPLASIA
(CAH)
Adreno– genital – syndrome

- It's **Autosomal recessive disorder (AR)** of the adrenal gland which involves a **deficiency** of an **enzyme** involved in the **synthesis** of cortisol, Aldosterone or both **(steroid biosynthesis)**

Incidence: about 1 in 15,000 births.

Enzyme deficiency:

1. **21 hydroxylase enzyme deficiency (the most common >90%).**
2. 17 Hydroxylase deficiency
3. 11 beta hydroxylase deficiency
4. 3 beta – hydroxylase deficiency



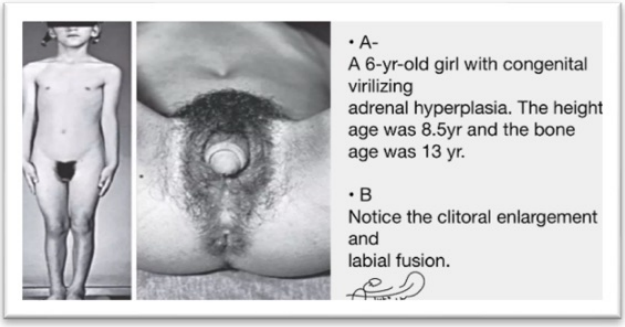
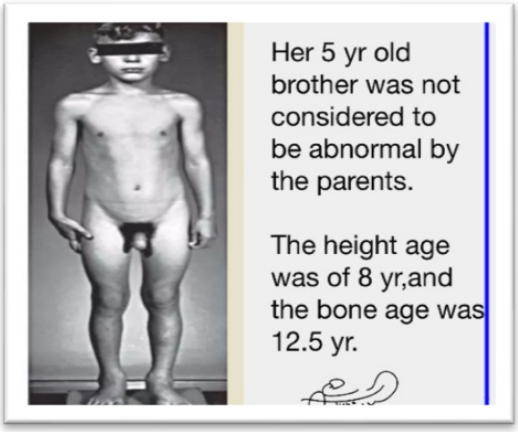
Congenital adrenal hyperplasia. Abnormal adrenal steroid biosynthesis from 21-hydroxylase deficiency is the most common form of congenital adrenal hyperplasia. ACTH, adrenocorticotrophic hormone.

Types of CAH: (21 hydroxylase deficiency)

1. **Classical type**
   - **Salt Wasting CAH** [Most Lethal form of 21 hydroxylase deficiency, Manifests as adrenal crisis at 2 to 6 weeks of life].
   - **Simple Virilising CAH**have milder defect with mineralocorticoid secretion and presented usually with precocious puberty]

2. **Non classical CAH** "Late onset"

Clinical picture:

Types	C/P in females	C/P in males
1–Salt Wasting CAH Sever form of CAH	• Ambiguous genitalia at birth (Increase Androgen in utero) • **Adrenal crisis.**	• My appear normal genitalia at birth • Big phallus – Hyper pigmented scrotum • 1st 1–3 weeks are usually present with a salt losing crisis (Adrenal crisis!!!) (V– D –weight loss, hypotonia– Dehydration – shock) (Low Na – high K – low Bp – low sugar – M acidosis)
2–Simple Virilizing CAH Milder Decrease in 21 hydroxylase	C/P in childhood precocious puberty • Precocious pubic hair + clitoromegaly • Accelerated growth & skeletal maturation (Tall Stature) Note (Adult Height is Short)  • A- A 6-yr-old girl with congenital virilizing adrenal hyperplasia. The height age was 8.5yr and the bone age was 13 yr. • B Notice the clitoral enlargement and labial fusion.	C/P in childhood precocious puberty • Early development of pubic hair &Penis enlargement • Accelerated linear growth & advanced bone age Note (Adult Height is Short)  Her 5 yr old brother was not considered to be abnormal by the parents. The height age was of 8 yr,and the bone age was 12.5 yr.
3.Non classical CAH "Late onset"	C/P in adolescence or adulthood 1. Oligomenorrhea 2. Hirsutism **acne 3. Infertility	
N.B	**Hypertension** is associated with **11 beta hydroxylase deficiency** B/C it increases 11 deoxycorticosterone which is powerful Na retaining steroids.	

Diagnosis:

A. History & examination (Antenatal History
   i. Family History (sudden neonatal death, or CAH)
   ii. General Examination & genitalia examination)

B. **Investigation:**

1. **Serum Level of 17 hydroxyprogesterone (very High)**
2. Urea & electrolytes Na (Low)
   i. K (High)
   ii. Urea (high) (If dehydrated)
3. Blood sugar (Low)
4. ABG: metabolic acidosis
5. 24 hrs Urinary 17 Ketosteroid (High)
6. **Karyotyping** (is essential in evaluation of any infant with ambiguous genitalia to establish the newborn sex)
   46XX ♀ 46 XY ♂
7. Bone Age (Advanced)

Management: needs multidisciplinary team

- **In case of adrenal crisis:**
  1. ABC
  2. IVF N/S 20 ml/kg rapid
  3. Hydrocortisone IV 50–100 mg/m^2 or 1–2 mg/Kg
  4. Dextrose D10% in hypoglycemia

- **Long Term treatment: (Lifelong hormone replacement therapy)**

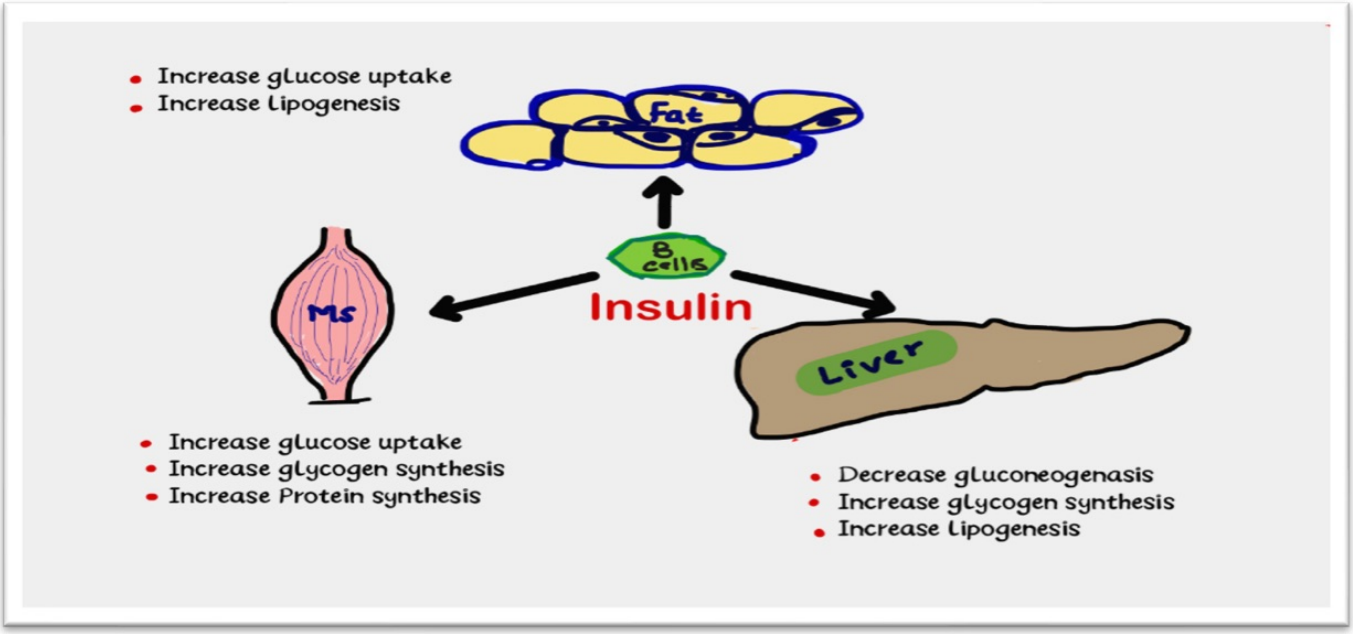
  1. **Hydrocortisone** orally (50 –100 mg/ m^2 / Day 6 hr)
  2. **Fludrocortisone**. (0.05 – 0.2 mg/day)
  3. Monitoring of growth, skeletal maturity (bone age), plasma androgens and 17α–hydroxy–progesterone is important in endocrine pod
  4. In case of infection – illness – surgery (Double the dose of hydrocortisone)
  5. Plastic surgery (corrective plastic surgery) & psychological Support

DIABETES MELLITUS

Definition:

Is a **chronic** illness characterized by **hyperglycemia** resulting from defects in insulin **secretion**, insulin **action**, or **both**.

DM is the most common endocrine disorder in pediatric "prevalence≃ 2–3/1000 children"



Classification of DM:

1. **Type 1 DM →** Most common in children (95%)
2. **Type 2 DM →** Usually in older children – obese– +ve F/H.
3. **Other Specific types:**
    1. Maturity onset diabetes of young **(MODY) →** Genetic defect in B–cell function +Strong family history.
    2. **Neonatal** diabetes: transient or permanent secondary to defective β–cell function.
    3. **Cystic Fibrosis** associated DM → pancreatic insufficiency
    4. Genetic / Chromosomal → **Down** – Turner's Syndrome
    5. Endocrine disease → **Cushing** syndrome
    6. **Drug**–induced, e.g. corticosteroids.

Features	Type I DM	Type II DM
Age of Onset	• Usually During childhood, puberty • Peak age (5–7 yrs– at the time of puberty)	• Commonly diagnosed over 10 yrs and at mid to late puberty.
Pathophysiology	• Absolute deficiency of insulin (destruction of pancreatic β–cells by an autoimmune process)	• Increase insulin resistance
Onset of C/P Family history of DM	Acute (rapid) → Days–wks • or +	• Gradual (months –yrs) + + + + (stronge)
Association	Other Autoimmune disease Hypo T_3 T_4 –Celiac disease–Addison.	• Obesity–PCO –Acanthosis nigricans –HTN–hyperLipidemia.
Autoantibodies	Positive ≃ 85% of cases {glutamic acid decarboxylase (GAD), zinc transporter 8 (ZnT8) and islet antigen 2 (IA–2)	Negative
DKA	Common Complication	Rare
Rx	Insulin Injection	Oral hypoglycemic– Exercise and diet.

Type I DM

Most children have type 1 diabetes (95%), although the number with type 2 diabetes is increasing.

- **Age**: Bimodal peak: (4–7 years) – (10–14 years)
- **Sex:** ♀=♂

Etiology: Multifactorial:

1. Genetic:

   A–Associated with HLA DR_3, DR_4
   B–Identical (Monozygotic) twins. (≃ 50–70 % Risk in 1st 3yrs)
   C–Familial risk: 2–3 % if affected mother
   - 5–6 % if affected father
   - 30% if both parents are affected

2. Environmental:
   - Geographical variation (more in Finland, Sweden)
   - Viral infection – increases Risk. (winter and autumn months)
   - Breast fed infants –Lower risk
   - Vit D deficiency

3. Autoimmune:

- Associated with other autoimmune disease hypo T_3 T_4 – Coeliac disease, rheumatoid arthritis.
- Positive Ab: Islet cell Ab – Glutamic acid decarboxylase Ab (GAD), zinc transporter 8 (ZnT8).

Clinical Picture:

Early: Classical Δ triad: Polyuria –Polydipsia –weight loss (Most common)

Less Common:

1. Enuresis (2ry).
2. Skin infection (candida).

Late: ≃ 30% of cases may present with DKA:

1. Abdominal pain (acute abdomen)
2. Vomiting –Dehydration–
3. Hyperventilation (Kussmaul Breathing) & Acetone smell
4. Drowsiness & Coma –hypovolemic shock.

Diagnosis:

1. **Symptomatic Child**
   - RBS ≥ 200 mg/dl
   - FBS≥ 126 mg/dl
   - 2hr postprandial blood sugar ≥ 200 mg/dl
2. **Asymptomatic Child**
   - At least 2 reading of blood sugar abnormal for 2 days
   - Oral glucose tolerance test OGTT (>200 mg/dl) → rarely required to diagnose type 1.

3. Urine glucose & ketone +++

4. Other Investigation:

   A. Glycosylated Hb (HbA_1C) (High >6 %):
      - Is used to assess & monitoring of glycemic control in the last 2–3 months
      - Is not helpful in case of hemoglobinopathies

   B. ABG if DKA is suspected –serum for ketone bodies.

   C. TFT & TTG (Screening for Hypo T_3 T_4 and coeliac disease) should be done annually.

Management of type I DM: needs diabetes team

Diabetes team includes:

1. Consultant paediatrician (s) with a special interest in diabetes(endocrinologist)
2. Paediatric diabetes specialist nurse(s)
3. Paediatric dietitian أخصائي تغذية
4. Clinical psychologist
5. Social worker
6. Parent/patient support groups

Goal of therapy (DM Management)

- Normal growth
- Emotional well-being
- Feels well
- Normal life & school performance.
- Control symptoms
- Prevent complications

- **Family Education:**

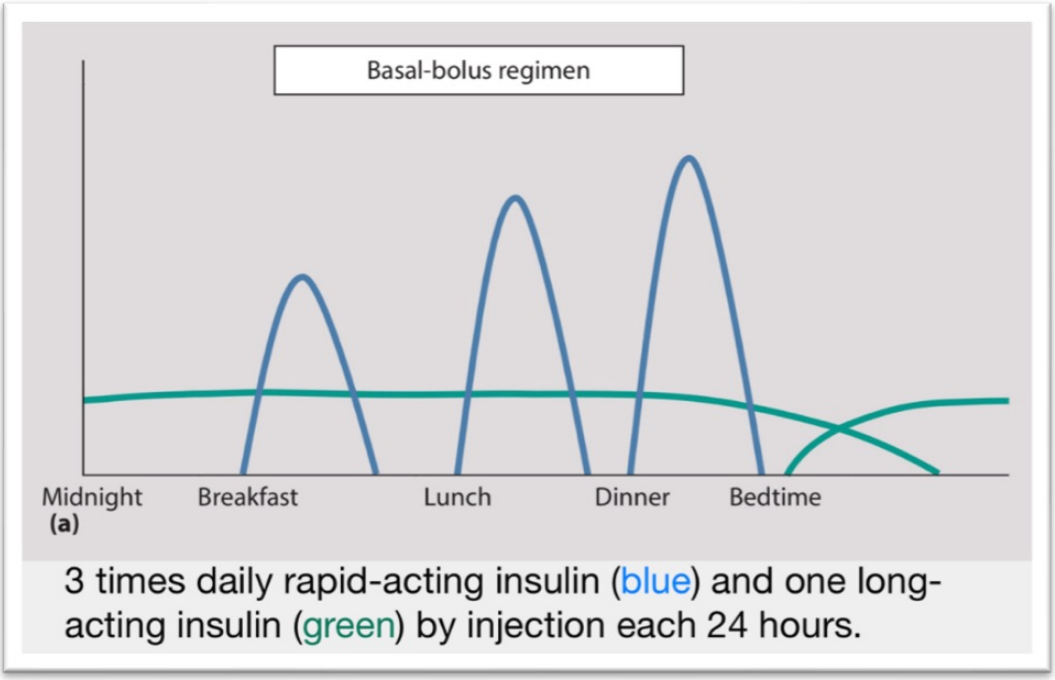
1. Basic understanding of pathophysiology of diabetes
2. Injection of insulin: technique and sites
3. Diet: regular meals & snacks – decrease refined CHO – healthy diet
4. Matching food intake with insulin and exercise
5. Sick – day rules during illness to prevent DKA
6. Blood glucose (Finger prick) monitoring
7. Recognition & Rx of hypoglycaemia
8. Psychological impact of life long condition & long-term complication

Insulin S C

- **Insulin:**
  - **Dose** (Prepubertal ≈ 0.5–1 U/kg/day)

    (During puberty: 1–1,2 u/kg/day (increase the dose upto 2 u/kg/day))
  - **Routes of injection** (Insulin syringe –Insulin pen–pump)
  - **Insulin regimens:**

    1. **Multiple daily injections** (basal– bolus) regimens (3–4 times /day)



3 times daily rapid-acting insulin (blue) and one long-acting insulin (green) by injection each 24 hours.

2. **Twice – daily regimen** (2/3 M & 1/3 E) not commonly used in pediatric diabetes.
3. **Continuous S C Insulin infusion (insulin pump)**: these regimens allow flexibility by relating the insulin more closely to food intake and exercise.

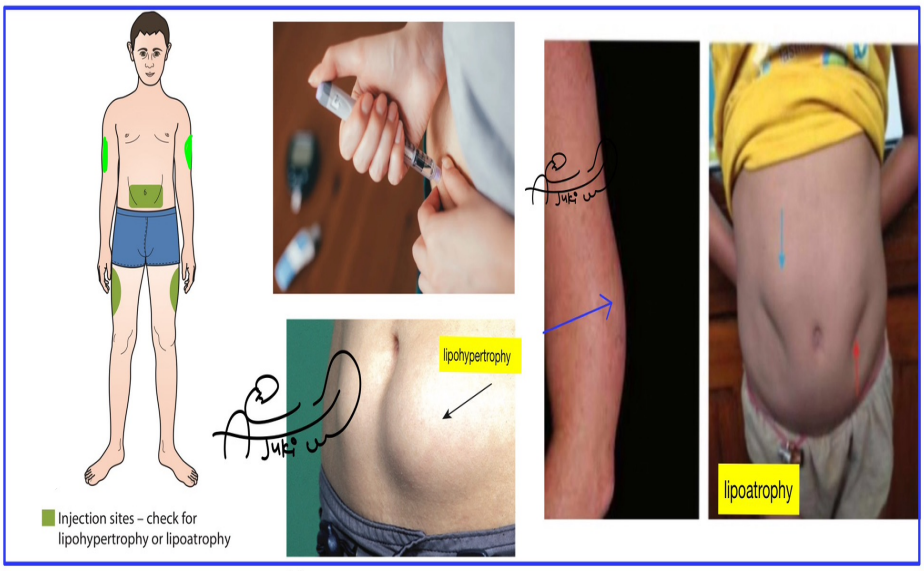
Insulin types:

Type of insulin	Onset	Peak action	Duration of action
1. Rapid acting insulin (lispro (Humalog), aspart (Novorapidبرتقالي), glulisine (Apidra)	10 – 20 min	1–2 hr	≃ 3–6hr
2. Short acting insulin(regular) Acta rapid – Human S	30 –60 min	2 – 4 hr	≃ 8 hr
3. Intermediate (NpH) acting insulin Exa Isophare I	1 – 2 hr	4 – 12 hr	≃ 18 hr
4. long–acting insulin analogues, e.g.detemir (Levemir اخضر),glargine (Lantus رصاصي), degludec (Tresiba فسفوري)	1–2 hr	8 –20 hr	up to 24 hours or more

Site of Insulin injection:

(outer arms – abdomen –thigh)

- True Injection **technique and rotation method are important** to prevent **lipohypertrophy**



(c)

Recognition & Rx of hypoglycaemia:

defined for someone with **diabetes mellitus as** a **blood glucose below70 mg /dl**

C/P:

(hunger, sweatiness, feeling faint, dizzy or 'wobbly' in their legs, hallucinations If unrecognized or untreated, symptoms may progress to loss of consciousness and seizures) Young children may be presented with pallor and a change in behavior.

Management:

(If suspected, the **blood glucose** level should be checked) **immediately**

- **If the child conscious:**
  (give him easily absorbed glucose in the form of glucose tablets or a sugary drink) (**role of 15** ***)

- **If the child is unconscious**:
  (treated either by IV dextrose 10% 2 ml /kg –or glucagon injection kit intramuscular)

- **Dietary management:**
  A healthy diet is recommended:
  A–complex CHO. (ovoid refined CHO)
  B– Low fat content
  C– Adequate protein
  D– High fibre

Carbohydrate counting: allows one to match the insulin dose to the choice of food, also taking into account the premeal glucose level and post meal exercise pattern.

- **Exercise:**

  It is an important aspect in management of DM and patients should be encouraged to exercise regularly.

  * Vigorous or prolonged planned exercise (cross-country running, long-distance hiking, cycling, skiing) requires **reduction** of the insulin dose and **increase** in carbohydrate intake.

- **Sick Day Management**

  • Insulin Management:

  – Insulin therapy must **always** be continued

  – Provide **extra** short acting insulin, if glucose is >300 or > trace ketones.

  – Estimate **10% – 15%** of total daily insulin dose.

- **Blood glucose monitoring:**

1. Regular blood glucose measurements are required to adjust the insulin doses and learn how changes in life- style, food and exercise affect glucose levels – Frequency (varies).

By using glucometer /Flash glucose monitoring systems/or Continuous glucose monitors (CGM)

2. HbA_1C (blood glucose monitoring in the previous 6–12 weeks)–target HbA1c of (< 7.0%)

Regular assessment of the child with diabetes (periodic):in diabetic opd visit (illustrated textbook)

• **growth** and **pubertal** development, avoiding obesity – measure height, weight and BMI and plot on growth chart **at each clinic visit**

. **Blood pressure** check for hypertension annually from **12 years** of age (age-specific centiles)

• **Renal disease** – screen for **microalbuminuria**, an early sign of nephropathy, annually from **12 years**

• **Circulation** – check **foot** pulses and **sensation**
• **Eyes** – retinopathy or cataracts are rare in children, but should be monitored annually from **12 years**, preferably with retinal photography

• **Feet** – maintain good **care**, avoid tight shoes and obtain prompt treatment of infections – **annually**

• **Screening** for **coeliac** and **thyroid** disease **at diagnosis**, thyroid screening **annually**, coeliac again if symptomatic

. **Annual** reminder to have **flu vaccination** and see dentist.

Complication of DM:

Acute complications	chronic complications
1. **DKA** 2. **Hypoglycaemia**	**1–macrovascular** – hypertension, coronary heart disease, cerebrovascular disease **2–microvascular**–retinopathy, nephropathy, neuropathy. **3–Complication of insulin therapy:** Lipodystrophy – (Lipoatrophy & Lipohypertrophy) Obesity Insulin Resistance **4–Mauriac syndrome:** (Type I DM +poorly uncontrolled) Growth failure, SS Delayed puberty Cushing face HS Megally

DIABETIC KETOACIDOSIS

- `≃ 30% of cases with Type 1 DM presented 1st time in DKA.

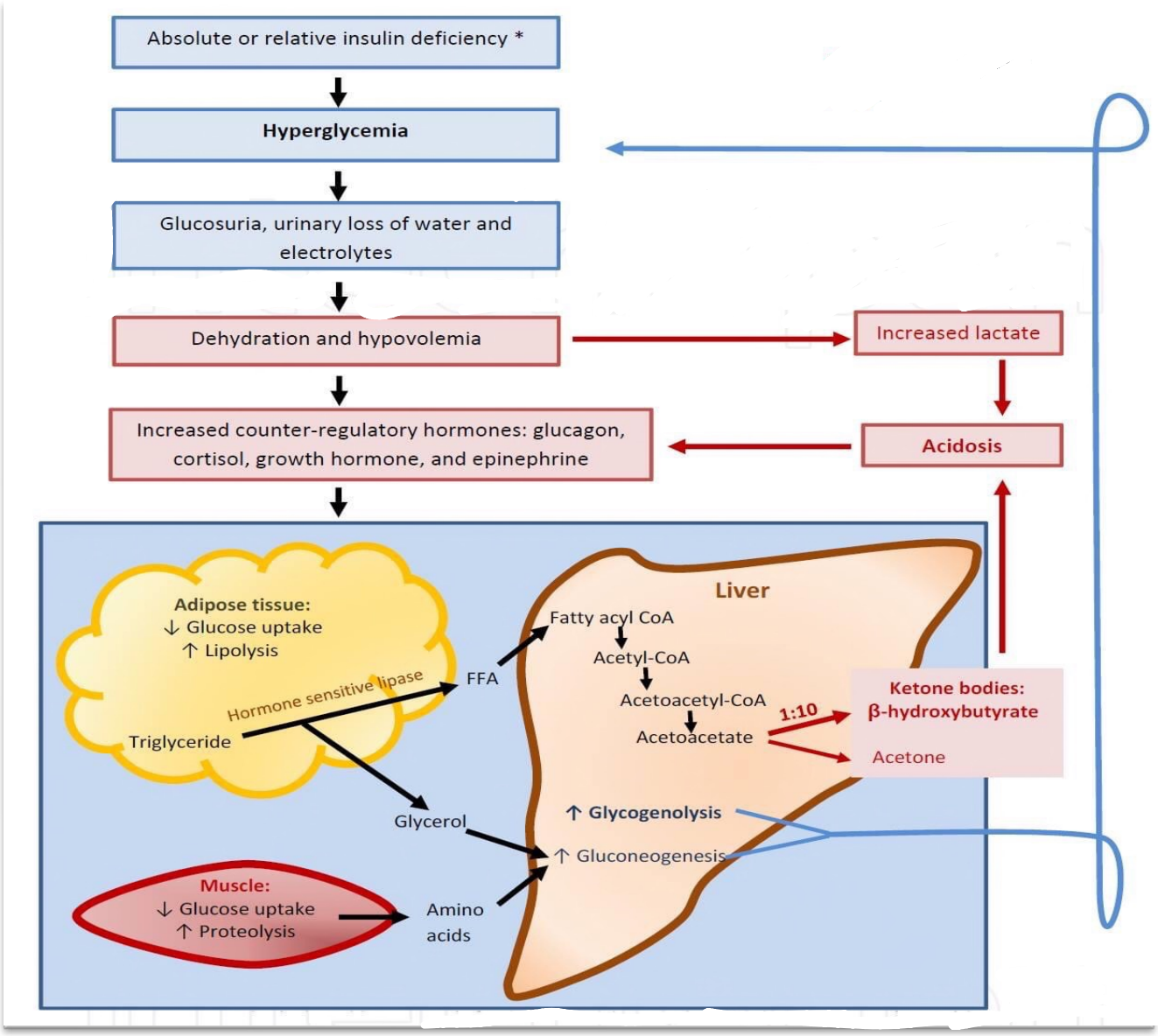
Causes (**Precipitating factors**) of DKA:

1. Omission of insulin dose.
2. Infection or stress.
3. Insulin Pump failure.

Diabetic keto-acidosis.

1. • Blood glucose >200 mg/dl.
2. • Ketones in blood & ketonuria.
3. • Acidosis (pH<7.3), HCO3 ≤15mEq/L.

Pathophysiology of DKA:



Clinical Picture:

- Early –polyuria + polydipsia – Nausea – Vomiting – Abdominal pain
- Then –Kussmaul Breathing – acetone smell – Dehydration
- Later –Confusion – Coma

Diagnosis:

1. RBS: >200 mg /dl
2. Urine for glucose & ketone bodies +++
3. ABG: Metabolic acidosis

Other Investigation: in case of DKA

- Urea and Electrolytes + K level
- Evidence of precipitating cause as infection (as blood or Urine C & S)

The Severity of DKA:

According to the 1-clinically 2-degree of acidosis

Degree of DKA	PH	HCO3
Mild DKA	<7.35	<15 mmol/L
Mild DKA	<7.25	<10 mmol/L
Severe DKA	<7. 15	<5 mmol/L

Treatment of DKA:

- **Goals of therapy:**
  - Correction of dehydration
  - Correct acidosis and reverse ketosis
  - Restore blood glucose to near normal
  - Avoid complication of therapy
  - Identify and Rx of precipitating factor

1. Good assessment of severity
2. **ABC, call for help & insert 2 live lines.**
3. Correction of **dehydration (fluids)**

If **shocked** → Rx of shock 10-20 ml/kg (Normal saline. 0,9%Nacl over 20 min)

Then:

Correction of **dehydration** (should be **gradually** over **48**-72 hr to **avoid cerebral edema**)

(M + D /over 48 hr)

- **Type of IVF** → Normal saline 0.9%Nacl
  → If the RBS ≤250 mg/dl **change** IVF to
  (½ D/S) or (5%dextrose+0.9%Nacl) to avoid hypoglycemia

ملاحظة مهمة (Don't decrease BL sugar Rapid >**100 mg**/dl/ hr)

Why ???

4. **Insulin:**

- Type Soluble (Regular) insulin
- Dose: 0.05 – 0.1 units/kg/hr
- Route: IV infusion (By infusion pump)
- Time: After one hour of rehydration ملاحظة مهمة

5. **Potassium (KCL):**

- Although initial serum k is high, it will fall following Rx with insulin and Rehydration
- Potassium replacement should start as soon as urine passed
- Add 20 – 40 m Eq/L of KCL in each 500 mL IVF
- Monitoring it by (ECG + ser K level)

6. **Acidosis Correction:**

- NaHCo3 should be avoided unless the child has very severe profound acidosis PH<7.1 & HCO3<5
- B/C NaHCO3 IV increase risk of cerebral edema ملاحظة مهمة

7. Identification and Rx of an **underlying cause** (antibiotic is not routinely prescribed in DKA)

8. **Monitoring**:

1. Fluid input –output chart.
2. ABG–blood sugar /regularly.
3. Electrolytes: Is especially **K** & Na (High glucose levels draw water by osmosis → This makes serum sodium appear low).
4. Neurological state assessment.
5. CVP & urinary catheter if shocked.
6. NGT if gastric dilatation – vomiting or deteriorating LOC.

- **Once the patient is improved** (**clinically + ABG**):

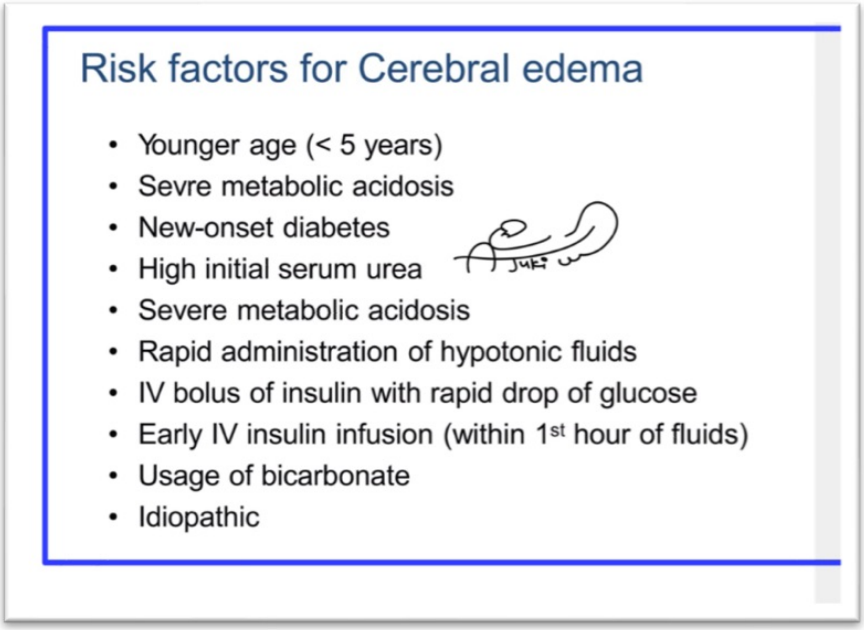
1. Stop IVF And start an oral diet.
2. Give S.C insulin every/6hr
   (ملاحظة مهمة: Stop intravenous insulin infusion 1 hour **after** giving subcutaneous insulin, as it takes time to be absorbed)

Complication of DKA:

1. Cerebral edema
2. Arrhythmia, electrolytes imbalance.
3. Acute tubular necrosis
4. Intracranial thrombosis
5. Pancreatitis, Aspiration Pneumonia.
6. DIC & multiorgan failure.

Cerebral edema:

Is the major life-threatening complication in DKA of unknown cause (% ≃ 1 %)
With MR 20-40%



C/P: S/S of increased ICP

- Headache.
- Vomiting.
- Cranial nerve palsy.
- Convulsion- Apnea – low HR-Cushing triad – confusion -Coma – Death

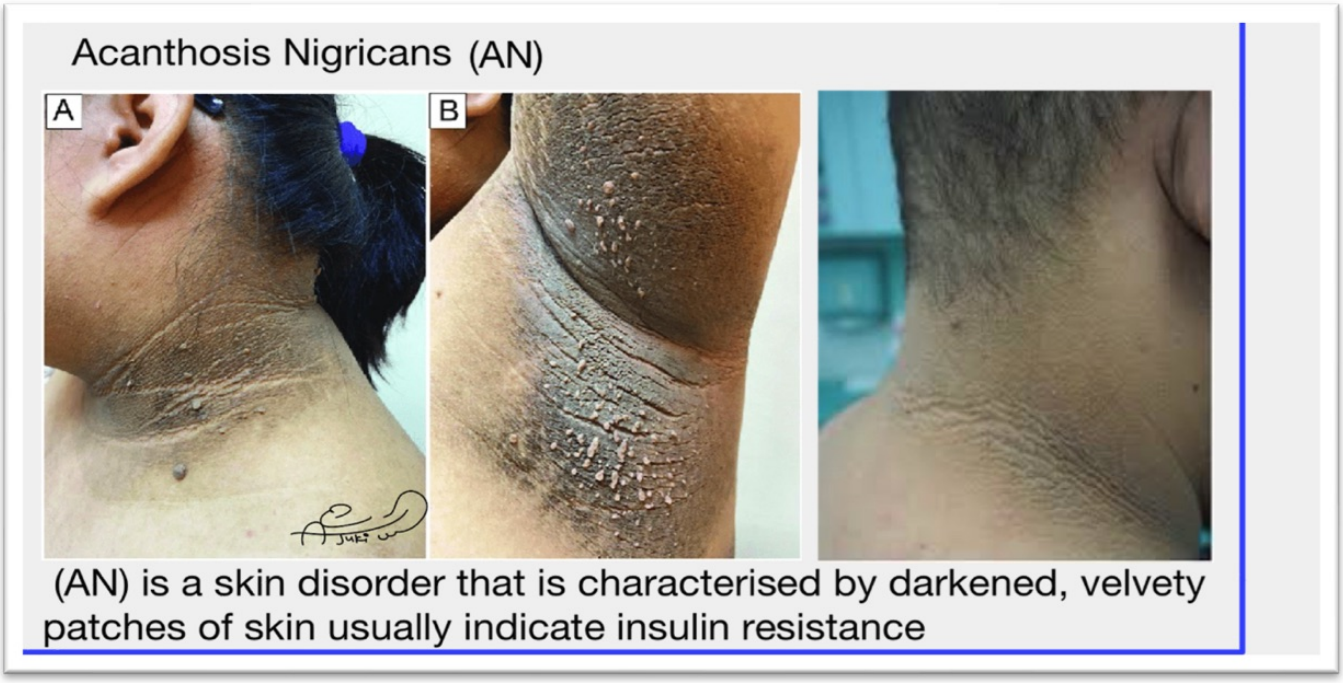
Treatment: PICU

- Decrease ICP → Semi Sitting position
  → mannitol IV 0.5-1 g/kg infused over 30 min.
  → Hypertonic saline – 3% IV infusion.
- ETT & MV

Honeymoon period	Somogyi Phenomenon	Down Phenomenon
• Occurs in type 1 DM (common 40–60% but Not all Pt) • Often suggestive remission • Some beta cells of pancreas have not completely destroyed & produce insulin → Pt become with Normal B1 glucose or recurrent hypoglycemia • The honeymoon phase can last for weeks, months or in some cases years.	• Normal glucose at night • High insulin given • Leads to hypoglycemia at around 2 –3 am • Increase glucagon & Anti insulin hormone • Lead to morning hyperglycemia ℞ decrease evening insulin ≃ 10%	• Normal glucose at night • Increase GH level at night • Morning hyperglycemia • ℞ Increase Evening insulin Dose ≃ 10%

Type 2 Diabetes in children & adolescents

- Most children with T2D are overweight or obese.
- More common in girls than in boys.
- Mild polyuria, polydipsia and little or no wt loss.
- DKA is less common than type 1.
- Usually diagnosed over 10 yrs and at mid to late puberty.
- Strong F/H of type 2 Diabetes (74–100 %)
- Evidence of insulin resistance, like acanthosis nigricans, PCOS, HPT, and hyperlipidemia.



Management of type 2 DM:

- Education
- Exercise and diet.
- Oral hypoglycemic drugs (**metformin**.).

EXAMINATION RELATED TO DM

Introduce self & explain what will do?

1. General Condition:

Look:

- Well – ill
- Weight & height – plot the reading on centile chart
- Color pale or pink
- Hydration state (sunken eyes)
- Type of breathing (Kaussmal)
- Surrounding IV line – IVF – infusion pump
- Odema
- Skin lesion hypopigmentation (Vitiligo)

2. Palpation (Specific Examination):

Hands & Upper limb:

- injection sites (lipohypertrophy) → (Arm – thigh – Abdo)
- Fingertip pricks (Glucometer site)
- Color of palmar creases!!
- Infection Between Fingers
- V/S pulse – Bp

Head & Face & neck:

- Eyes – pallor – VA
  Cataract – Exa of Fundus
- Mouth – Color of mm – Acetone smell
  Mouth hygiene
- Neck – Thyroid swelling
  Acanthosis nigricans

Abdomen:

- Injection **sites** – Distension
- Hepatomegaly
- Tanner staging

Lower limbs:

• Injection sites • Infection between toes • Sensory examination	

CHILDHOOD HYPOGLYCEMIA

is **defined** as a blood glucose less than 2.6mmol/L= 45 mg /dl (in children without diabetes mellitus). " Illustrated textbook "

Causes:

Insulin excess	Without hyperinsulinemia
1. Excess exogenous insulin, e.g. in diabetes mellitus, maternal diabetes.	1. Ketotic hypoglycemia of childhood
2. β–cell disorders – persistent hypoglycemia hyperinsulinism of infancy, insulinoma	2. Inborn errors of metabolism, e.g. glycogen storage disorders, galactosaemia, fatty acid oxidation defects
3. Drug–induced (sulphonylurea)	3. Hormonal deficiency: GH↓, ACTH↓, Addison disease, congenital adrenal hyperplasia
4. Autoimmune (insulin receptor antibodies)	4. Drugs – alcohol, aspirin, beta blockers
5. Beckwith–Wiedemann syndrome	5. Sepsis –Liver disease

Idiopathic ketotic hypoglycemia

- The most common cause of hypoglycemia in children (age 1–5 year)
- Peak Age (18 – 36 months)
- Etiology: Unclear (probably due to limited reserves for gluconeogenesis)

C/P:

- Usually presented in the morning (episodes of hypoglycemia)
- Neuroglycopenia – lethargy – malaise – Headache
  Confusion – visual disturbances
  Dizziness – Seizure – coma

- The child usually slim & underweight + (short – thin)
- Urine for ketone (positive)–& insulin levels are low

R_x

- Parents advised to avoid prolonged fasting
- This will resolve by time

Neonatal hypoglycemia

Definition:

serum sugar less than 45 mg /dL. Is common problem in NICU
WHY IS IMPORTANT: (B/C severe persistent or recurrent hypoglycemia may lead to brain damage –learning impairment– & epilepsy)

Causes:

Hyperinsulinism	Decreased production /stores	decreased utilization / Decreased production	endocrine disorders
1–diabetic mother 2–Beckwith–wiedemann syndrome 3–hydrops fetalis 4–LGA 5–maternal drugs as propranolol 6_islet cell hyperplasia	1–Preterm 2–IUGR 3–Inadequate calorie intake	1–Perinatal stress 2–sepsis –asphyxia–shock – hypothermia 3–exchange transfusion – Polycythemia 4–metabolic diseases (GSD –Fatty acid disorder– galactosemia–organic acidemia – tyrosinemia)	1–CAH (adrenal cri 2–hypopituitarism (GH deficiency)

Investigation:

1– blood glucose (Glucometer –Lab)
2– If the cause of the hypoglycemia is unknown: urgent sample of blood.

Blood sample:

- Confirm hypoglycemia with laboratory blood glucose

- Growth hormone, cortisol, insulin, C-peptide, ketones (acetoacetate, B hydroxybutyrate), bicarbonate, lactate, ammonia, amino acids, acylcarnitine profile.

First urine after hypoglycemia:

- Organic acids
- Consider saving blood and urine for toxicology, e.g. salicylate, sulphonyl urea

Treatment: **ABC**DEFG (((**DEFG** → ('**D**on't **E**ver **F**orget **G**lucose').

***Hypoglycemia can be corrected with an intravenous infusion of glucose (**2–5 ml/kg of 10%** glucose bolus followed by 10% glucose infusion) or orally if stable and conscious.

***Other drugs can be used (Glucagon IM –corticosteroids)

DIABETES INSIPIDUS

Definition:

is a disorder of the posterior lobe of the pituitary gland characterized by a deficiency of antidiuretic hormone (ADH) = arginine vasopressin, Great thirst (polydipsia) and large volumes (polyuria) of dilute urine (<300 mosm/kg) characterize the disorder.

1. Central DI (Neurogenic)	2. Nephrogenic DI
Decrease ADH secretion from **post pituitary**	**Resistance** to ADH action in the **kidneys**
Causes:	**Causes:**
1. Head Trauma or post-surgery 2. Brain Tumor as craniopharyngioma 3. Idiopathic ≈30% 4. CNS Infection 5. Congenital(hereditary)–wolfman syndrome	1. Congenital (XLR) 1ry DI 2. Drugs: Lithium 3. Electrolyte disturbances (hyper Ca – hypo mg – hypo K) 4. CRF – interstitial nephritis–SCA.

C/P:

1. Polyurea
2. Polydipsia

3. FTT– constipation.
4. 2ry Nocturnal enuresis
5. Dehydration – shock in severe cases
6. In Infant (crying – irritability – FTT –fever)

Diagnosis:

1. 24 hr urine collection for determination of urine volume (high)
2. Serum osmolarity >300 mosm/kg
3. Urine osmolarity <300 mosm/kg
4. Serum Electrolyte: Na (High)** –ca –k
5. Water deprivation test & response to DDAVP can → used to confirm diagnosis & to differentiate between central & nephrogenic DI
6. MRI brain to determine underlying cause of central DI

IN A NORMAL WELL HYDRATED PATIENTS (PLASMA OSMOLALITY IS<300 mOsml/I &. URINE OSMOLALITY IS 300–450 mOsmol/I)

Treatment: in addition to adequate fluids & calorie intake

1. Central DI: Desmopressin (DDAVP)
2. Nephrogenic DI: Thiazide diuretics & chlorpropamide naka– indomethacin

Polyurea

- Is defined as an increase in total daily output of urine.
- Urine output → **> 40ml/kg/24hrs** or **>2000 ml /m2/24hrs**
    - Preschooler children **>** 1 L/24hrs
    - School children >2 L/24 hr & Adults >3 L/24hrs)

Causes of Polyuria & polydipsia in children:

1– DM. 2– DI 3– primary polydipsia.

4– Renal causes (chronic kidney disease —Renal tubular acidosis—Fanconi syndrome—Bartter syndrome)

5– Hypercalcemia & hypokalemia

6_ Drugs: I. Diuretics. II. Vitamin D. III. Lithium. IV. Amphotericin B

SHORT STATURE

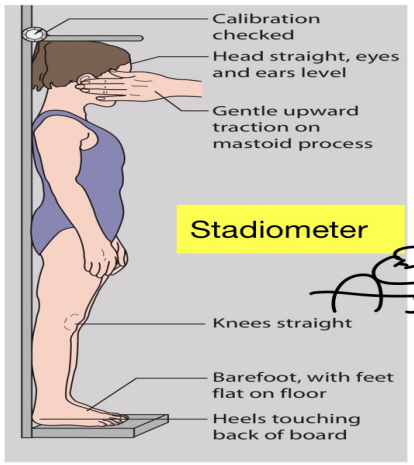
Definition:

Short stature is usually defined as a height below the second centile (i.e. 2 SDs below the mean)

Also defined as height is below 3rd centile for age & sex (WHO chart)

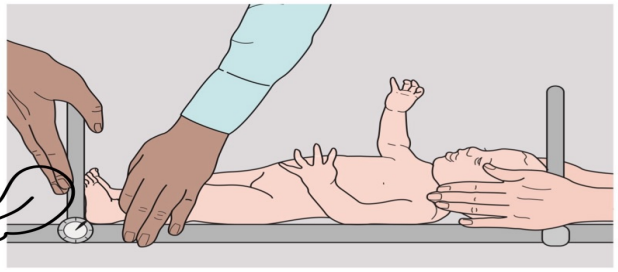
N.B:

Length: Is measuring the infant in supin position (<2 years)
Height: Is measuring the child in standing (>2 years)



Measuring height in children.

Infantometer



Measuring length in infants and young children. An assistant is required to hold the legs straight.

Normal Length at birth = 50cm
Normal Length at 1 year = 75 cm
Normal Length at 4 years = 100cm

Height Formula > 2 Years = Age in years x 5 + 80

Upper Segment: Lower Segment Ratio

- Lower segment = Distance between ASIS & Floor
- Upper segment = Setting Height = Height – Lower segment

Normal U/L segment Ratio at birth = 1.7 :1
Normal U/L segment Ratio at 3 years = 1.3: 1
Normal U/L segment Ratio at 7–10 years = 1:1
Normal U/L segment Ratio at puberty = 0.9 : 1

Growth Velocity:

It's sensitive indicator of growth failure

(How many cm gained / year) → 1st year 23– 27 cm /year
(How many cm gained / year) →2–5 years ≥6cm / year
(How many cm gained / year) →5 year – puberty 5–6 cm / year

Mid Parental Height:

Boys = (Father height mother height) /2 +6.5
Girls = (Father height mother height) / 2 –6.5
Then plot the result on Growth Chart at Age 20 to form Family chart (–+ 8,5 10 cm)to form centile.

Types

Proportionate SS

Normal variants *

1-Familial SS* mc

2-Constitutional SS

Pathological

1-Chronic diseases

2-Endocrine disease

3-Genetic (chr)

4-Nutritional deprivation

5-Psycho-social

6-IUGR

7-Drugs

Disproportionate SS

é Short limbs

Achondroplasia

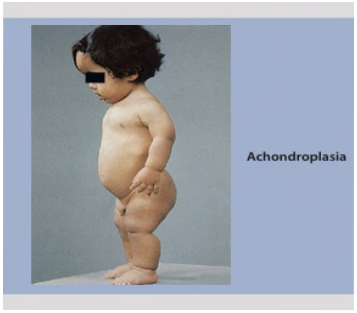
Osteogenesis imperfects

Rickets

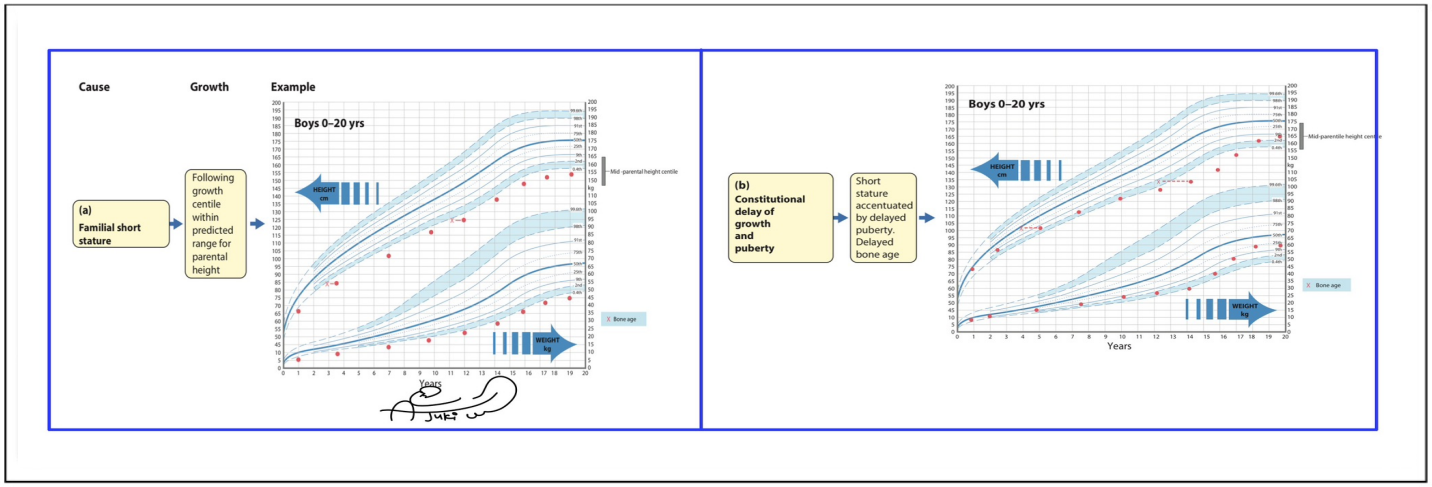
éShort trunk

Kyphoscoliosis

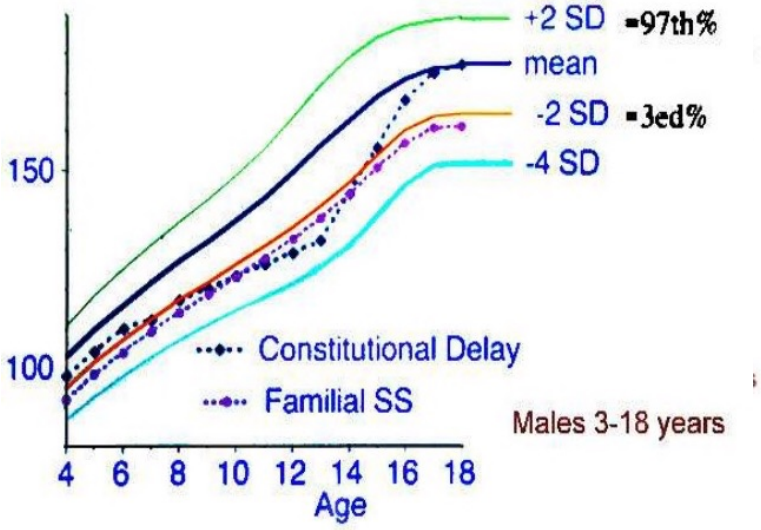
MPS



Features	Familial SS	Constitutional Growth delay & puberty (CGDP) "Late bloomer"
✓ At birth	✓ Small birth length	✓ Normal Newborn
✓ F/H of SS	✓ Positive (yes)	✓ Negative (No)
✓ Gender	✓ (♀=♂)	✓ (more in ♂)
✓ F/H of delayed puberty	✓ Negative (no)	✓ Positive (yes)
✓ Puberty	✓ Normal	✓ Delayed
✓ Height velocity	✓ Normal	✓ Normal
✓ Bone Age	✓ BA=CA	✓ BA (delayed)
✓ Adult Height (Final Height)	✓ Short	✓ Normal



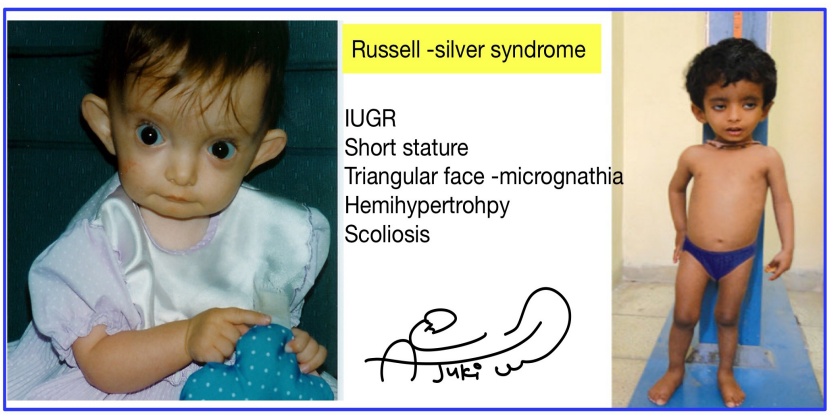
Pathological SS:



1. **Endocrine causes:**
    2. Hypothyroidism
    3. GH deficiency (Hypopituitarism)
    4. Cushing syndrome
    5. Pseudohypoparathyroidism

2. **Chronic Disease:**
    1. **CVS**: CHD
    2. **Respiratory**: Cystic Fibrosis
    3. **GIT**: Malabsorption as (celiac disease – IBD)
    4. **Blood**: Chronic hemolytic anemia as thalassemia
    5. **Renal**: Chronic kidney disease
    6. **chronic inflammation as** juvenile idiopathic arthritis

3. **Chromosomal Disorder / Syndromes:**
    1. Down syndrome (Trisomy 21)
    2. Turner & Noonan's syndrome
    3. Williams syndrome
    4. Prader willi syndrome
    5. Russell Silver syndrome



4. **Nutritional deprivation:** As marasmus

5. **Psychological deprivation:** Physical & emotional deprivation may lead

To growth failure & delayed puberty

6. **Intrauterine growth restriction (IUGR) & Extremely Prematurity:**

≃ 1/3 of cases remain short

7. **Drugs:** (Excess steroid)

8. **Metabolic Disease:** As mucopolysaccharidosis

Important definition:
• **Chronological age:** Actual age of the child (age since birth) • **Height age:** it's the age at which the height of the child is at 50th centile. • **Bone age**: is an indicator of skeletal maturation.

Investigation:

1. **Bone age** (left wrist & hand X Ray) either normal or delayed.
(Greulich–Pyle Atlas method (GP method) most widely used systems)

Causes of Delayed Bone Age:	causes of Advanced bone Age:
1. Hypo T_3 T_4 2. Hypopituitarism 3. Cushing Syndrome (Cortisol excess) 4. Delayed Puberty 5. Chronic illness – –Undernutrition	1. Hyper T_3 T_4 2. Simple Obesity 3. Congenital Adrenal hyperplasia 4. Precocious puberty 5. Hyperpituitarism

2. **Other Investigation:**
   - CBC – Anemia celiac – CKD
   - Cr /Urea – CKD
   - TTG –LFT
   - Hormonal Assay – 1. TFT: hypothyroidism
     – 2. IGF-1
   - Karyotyping – If their dysmorphic features

- Other **special** investigations (ESR –CRP –ser ca, ph, ALp – vit d –MRI brain–GH provocation test –Echo –immunoglobulin level)

Treatment of SS: mainly R_x of the underlying cause

1. Counselling of parents (for familial or CGDP SS)
2. Dietary advice as in (Undernutrition, Celiac disease)
3. Growth hormone therapy inj S C / Daily used in some cases as mentioned in the HYPOPITUITARISM lecture. (Indication of GH therapy حفظ)

Causes of tall stature (Excessive growth) mcq

1. **Familial** (Constitutional) – most common cause
2. Idiopathic **Obesity**
3. **Secondary**
   - Increase T_3 T_4 – Hyperthyroidism
   - Increase Androgen – CAH
   - Increase GH – true Gigantism
   - Precocious puberty
4. **Syndromes**:
   - Marfan's Syndrome
   - Homocystinuria (metabolic disease)
   - Klinefelter's Syndrome (47 XXY)
   - Beckwith Wiedemann Syndrome
   - Sotos Syndrome (associated with **large** head, characteristic facial features, and learning difficulties)

PUBERTY

Definition:

Is a period of life at which the child sexually, physically, and psychologically becomes mature (appearance of 2ry sexual characters)

TERMS: –

- **Pubarche**: appearance of pubic hair.
- **Thelarche**: appearance of breast tissue.
- **Menarche**: age of the first menstrual period.
- **Spermarche**: age of first ejaculation.

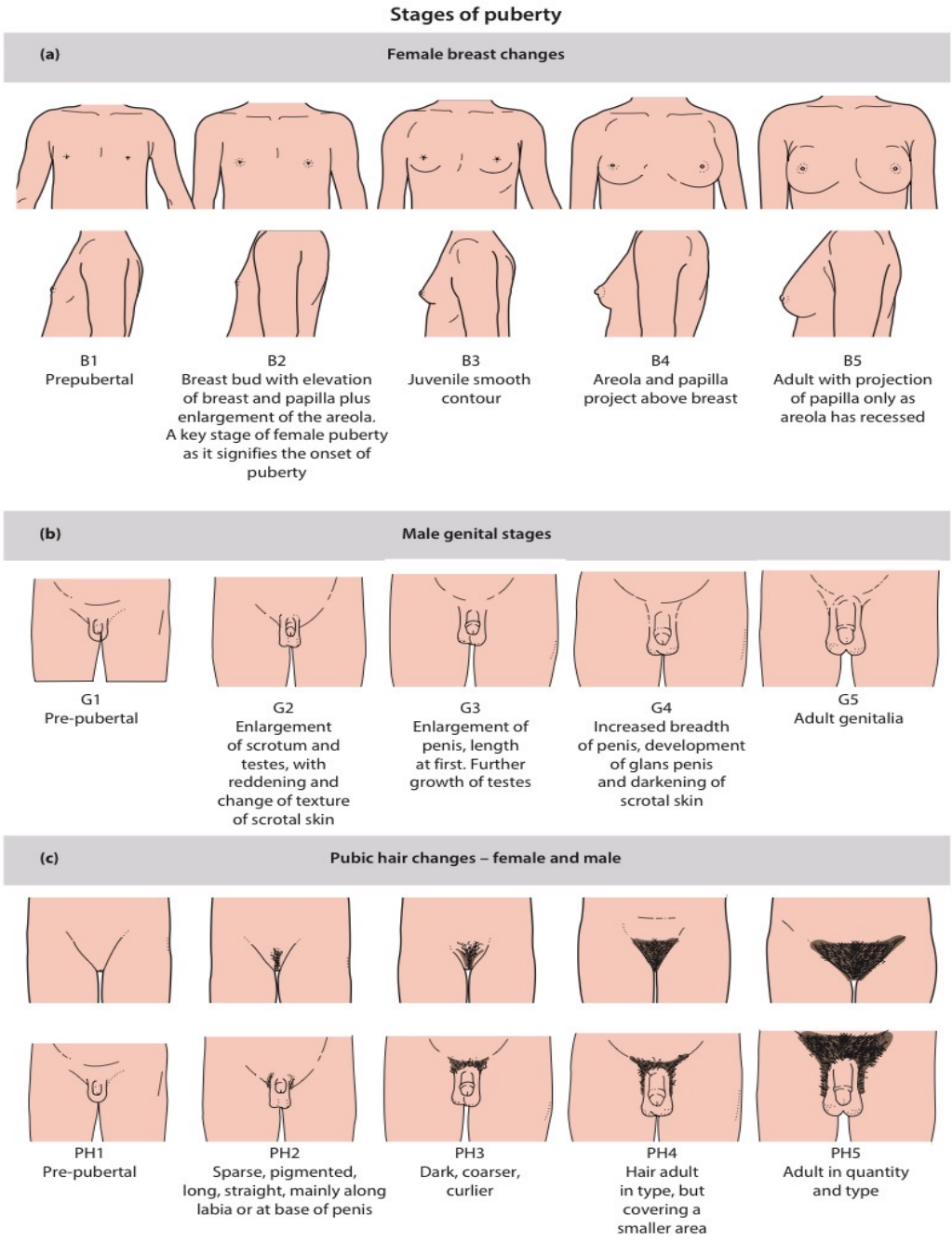
Features of Puberty:

	Female ♀	Male ♂
Onset	• **(8–13 Years)**	• **(9–14 years)**
1st sign of puberty	**Breast Enlargement (Thelarche)** = breast Stage 2 Followed by →Pubic hair–>axillary hair → Menarche	**Testicular Enlargement** = Testes 4ml (by orchidometer) Followed by Pubic hair –>thinning & pigmentation of the scrotum →incr Penis size → axillary hair
Puberty Growth Spurts	Occurs Immediately after breast enlargement (**Early**)	Occur **late** after ≈ 18m of testicular Enlargement

Note:

- In both sexes, there will be development of acne, axillary hair, body odor, and mood changes.
- Menarche (in girls) is the last sign of Puberty in ♀ (occurs on average 2.5 years after the onset of puberty)
- 1st seminal discharge (in boys) is the last sign of Puberty in ♂

Tanner staging of puberty: also known as Sexual Maturity Rating "SMR"



Schematic drawings of male and female stages of puberty. Pubertal changes are shown according to the Tanner stages of puberty.

Orchidometer to assess testicular volume (in ml). (From: Wales JKH, Rogol AD, Wit JM: *Pediatric Endocrinology and Growth*. London, 2003, Saunders, with permission.)

Prader orchidometer

Disorders of puberty

Either **precocious** puberty **or delayed** puberty

1. Precocious puberty

Definition:

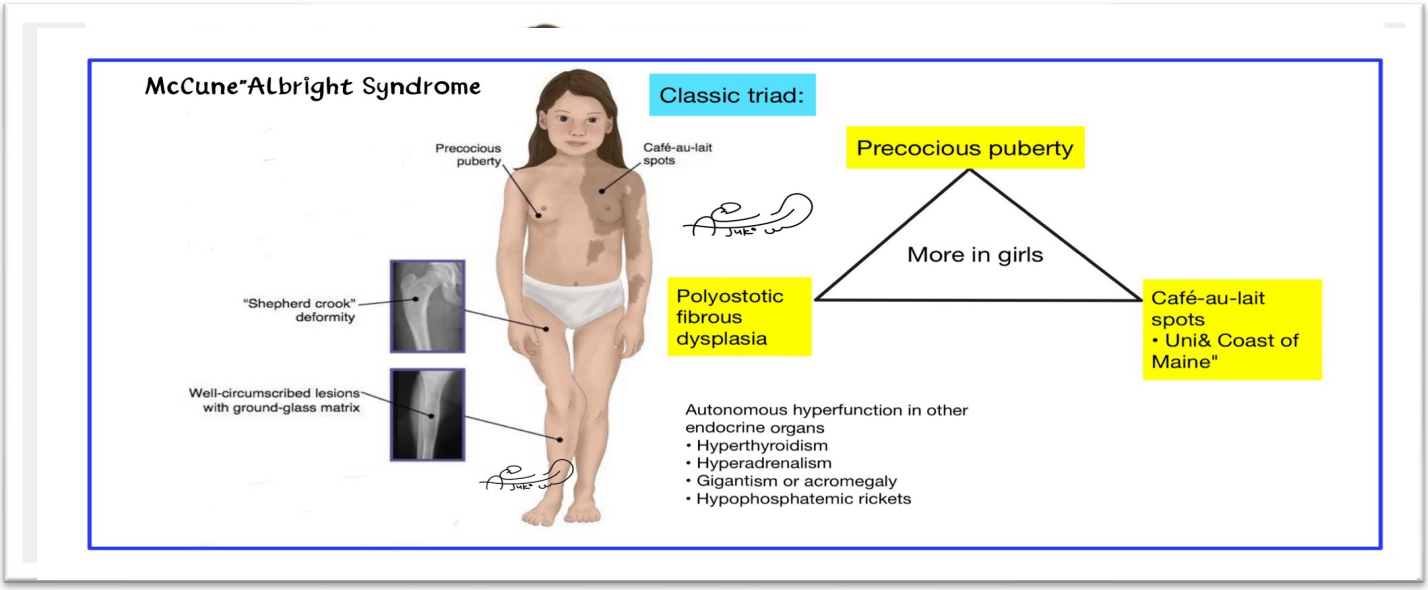
Development of Secondary sexual character < 8 Years in ♀ & <9 years old in ♂.
It may be due to:

1. **Precocious puberty** (True pp or pseudo pp)
2. **Premature breast enlargement (thelarche)**
3. **Premature Public hair development**

Central (true) PP	Pseud (false) PP
• Gonadotropin – dependent (**high** LH, FSH) • signs of puberty are **consonant** • due to **premature** activation of the hypothalamic–pituitary–gonadal **axis**.	• Gonadotropin independent (**Low** LH, FSH) • signs of puberty often **not consonant** • due to **excess** sex steroids produced **outside** the pituitary gland.
Causes: 1. **Idiopathic** (constitutional, functional) (Common in ♀) 2. **CNS abnormalities**: A–Congenital anomalies, e.g. hydrocephalus B–Acquired, e.g. post–irradiation, infection, surgery, brain injury C–**Tumours**, e.g. craniopharyngioma, neurofibromatosis 3. **Hypothyroidism**	**Causes:** 1. **Adrenal** disorders – tumors, congenital adrenal hyperplasia 2. **Ovarian** – tumor (granulosa cell) 3. **Testicular** – tumor (Leydig cell) 4. **Exogenous** sex steroids 5. **McCune**– Albright Syndrome

NB:

- PP in **Females ♀** is **common** & usually (familial –Idiopathic) Physiological due to premature onset of normal Puberty.
- PP in **males ♂** is **rare** & usually abnormal & **pathological** causes must be excluded



Investigations:

After good Examination & pubertal staging.

- Bone Age
- U/S pelvis (girls)
- LH & FSH level – Estrogen & Progesterone or testosterone level
- MRI pituitary gland & brain

Treatment:

It depends on the underlying cause

- True PP: Rx by GnRH analogues
- False PP: ♂ Androgen inhibitors
  ♀ Estrogen inhibitors

1. Isolated Premature breast Enlargement (thelarche): مهمة	2. Premature Adrenarche(pubarche)
• Usually affect ♀ between 6m & 2 years of age • Breast Enlargement may be asymmetrical (usually not beyond stage 3)	• Development of pubic hair <8 in ♀ & <9 years in ♂ • No other signs of sexual development, significant growth acceleration • Common in Asian & Afro-Caribbean children • Due to androgen production by adrenal glands • Usually self-limiting

• No other sign of puberty or growth spurt • It's benign condition – Non progressive & self-limiting • May be associated with ovarian cyst by U/S. Ix: not required usually Rx: Reassurance & follow Up	• But U/S – BA should be obtain to exclude central PP. • CAH should be excluded • Follow Up is essential

Delayed Puberty

Definition:

- The **absence** of pubertal development by **13 years** of age in ♀ & **14 years** in males.
- Delayed puberty is **common** in **boys** ♂ & the cause is usually **Constitutional delay of growth and puberty /familial.**

Causes:

Low Gonadotropin secretion	High Gonadotrophin secretion
1. Hypothalamo – Pituitary disorders: • Panhypopituitarism • GnRH deficiency • Kallmann Syndrome • Intracranial tumor 2. Systemic Disease CF, CKD, Crohn disease, organ dysfunction, anorexia nervosa, starvation, excess physical training 3. Acquired Hypo T_3T_4	1. Chromosomal Abnormalities • Turner Syndrome (45XO) • Klinefelter Syndrome (47XXY) 2. Gonadal Disease • Post-Surgery • Post Trauma radiotherapy or-Chemotherapy

N.B

_ **Delayed puberty in girls** is **less** common and a **pathological** cause should be **excluded**.

_ **Karyotyping** should be performed for **Turner** syndrome, and **thyroid** and **sex** steroid hormones should be measured. Pituitary **pathology** should be excluded by an **MRI** scan.

Treatment:

1. Identify & treat any underlying pathology
2. Accelerate growth & promote entry into puberty if necessary (as estradiol for several months to induce puberty in girls).

PARATHYROID GLAND DISORDERS

Ca Homeostasis:

This is controlled mainly by:

1. PTH
2. Vitamin D
3. Calcitonin

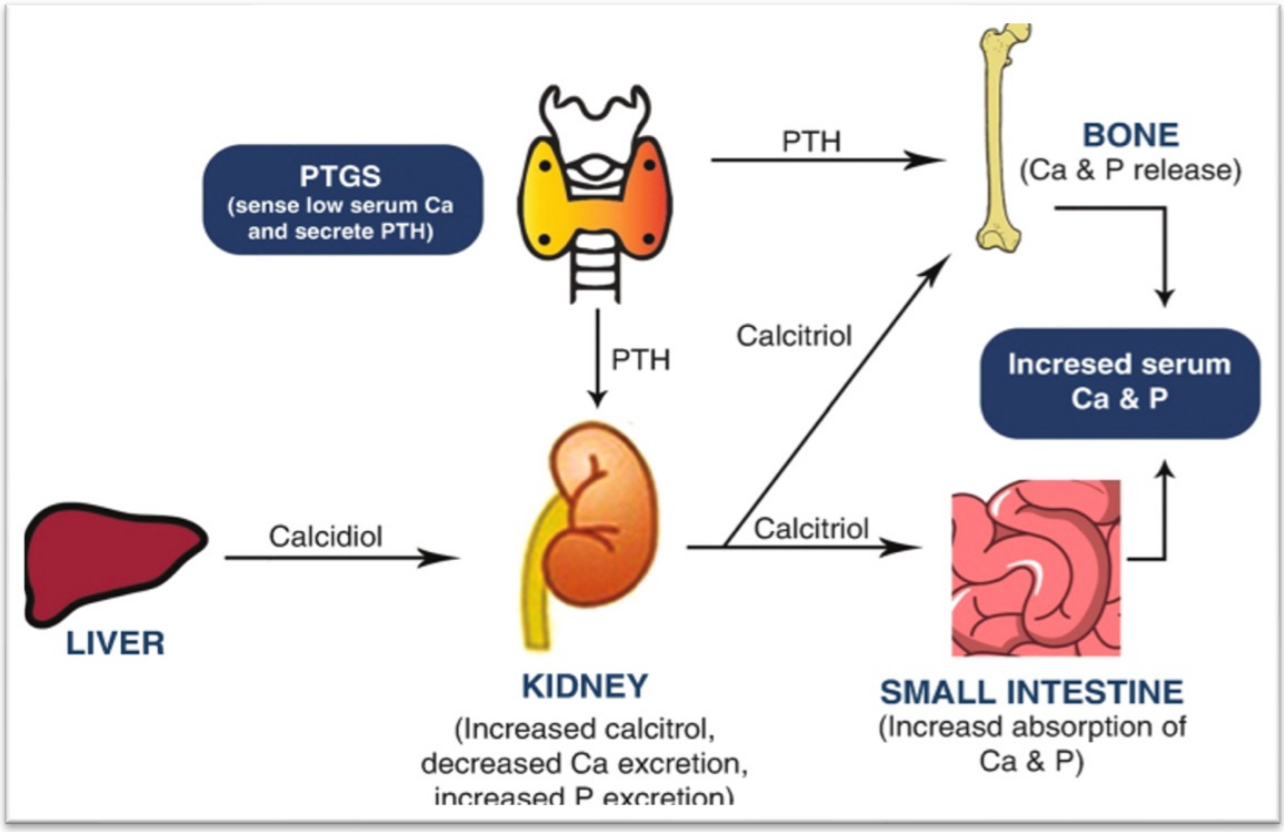
Blood Calcium

A) **Non-diffusible** (45%): Bound to albumin & globulin. It is physiologically inactive

B) **Diffusible**:

a. Non-iodized (5%): Forming complexes, with citrate, bicarbonate, POa

b. Ionized (50%): It is the physiologically active form.



Hypoparathyroidism

- Is rare in childhood

Causes:

1. Congenital causes – Digeorge Syndrome
2. Iatrogenic (after thyroidectomy)
3. Autoimmune (Autoimmune poly glandular syndrome I)
4. Transient neonatal hypo function (Common in Preterm, IDM, Asphyxia)

Clinical Picture: (Hypocalcemia) ca level<8.5 mg/dl

1. **Manifest** tetany (Paresthesia – ms cramps – Hyperirritability –Carpopedal spasm – Hoarseness– stridor – seizures)

2. **Latent** tetany: diagnosed by provocative tests
   - Chvostek's sign: Tapping of the Facial nerve → contraction of facial ms
   - Trousseau's sign: Compression of the upper arm → carpal spasm.

Investigations:

1. Serum Ca (Low), Ph (high), ALP (N or Low)
2. PTH Level (Low)

Rx:

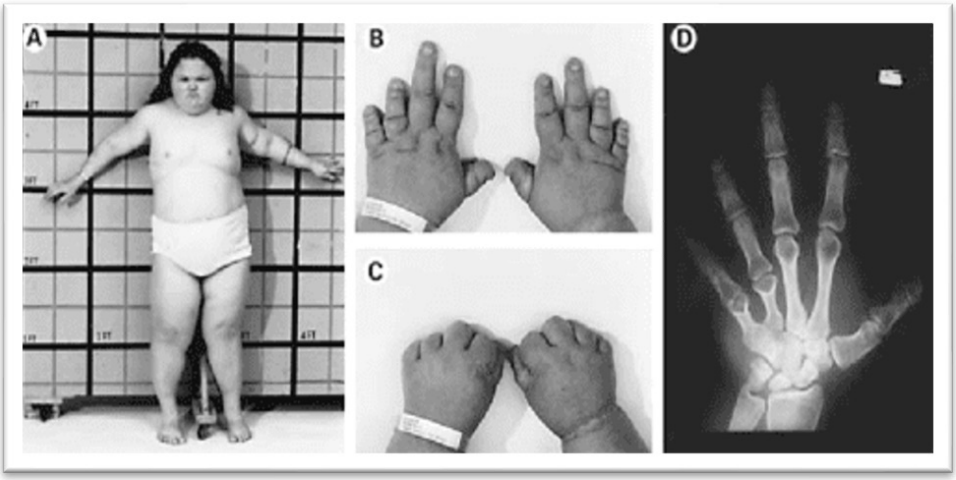
1. Correct hypo Ca by (Ca Gluconate IV slowly if severe) then orally
2. Vitamin D supplement

Pseudohypoparathyroidism (Albright hereditary Osteodystrophy)

- AD inheritance, there is **end organ resistance** to the action of PTH caused by mutation in signaling Molecules in the receptors.
- Serum Ca (Low)– Serum Ph (High) but **PTH (high)**

C/P:

- Short Stature – Round face
- Stocky habitus (obese)
- Short 4th & 5th Metacarpal bone – Dimple Sign**
- Intellectual Disability (MR)
- Calcification of the basal ganglia



Pseudo Pseudo hypoparathyroidism

- C/P of Pseudo hypoparathyroidism
- Normal Serum Ca – Ph.

Other Causes of Hypo Ca: (Infant & children):

1. Vitamin D deficiency
2. CKD
3. Hypo mg –hyper ph.
4. Malabsorption
5. Low serum Albumin
6. Liver Disease, Tumor Lysis syndrome –massive blood transfusions

Hyperparathyroidism (rare in children)

1. **Primary:** Single Adenoma – Familial
2. **Secondary:** CKD – Liver Disease
3. **Tertiary:** In case of Long-Standing CKD.

C/P of hypercalcemia:

1. Asymptomatic.

2. symptomatic: They are classically summarized by the mnemonic "**stones, bones, abdominal groans and psychic moans**"

1. (**Stones**) refers to kidney stones, nephrocalcinosis, and diabetes insipidus (polyuria and polydipsia).
2. (**Bones**) refers to bone-related complications, osteitis fibrosa cystica, which results in pain and sometimes pathological fractures, osteoporosis, osteomalacia, and arthritis.
3. (**Abdominal Groans**) refers to gastric related symptoms such as Abdominal pain, vomiting, dehydration, constipation and anorexia

4. **Psychic moans**" which includes depression, loss of memory –psychosis ataxia –Confusion– coma

Ix:

1. Serum Ca High (>12 mg/dl)
2. Serum Ph Low (<3 mg/dl)
3. PTH Level High

Rx of Server Hyper Ca:

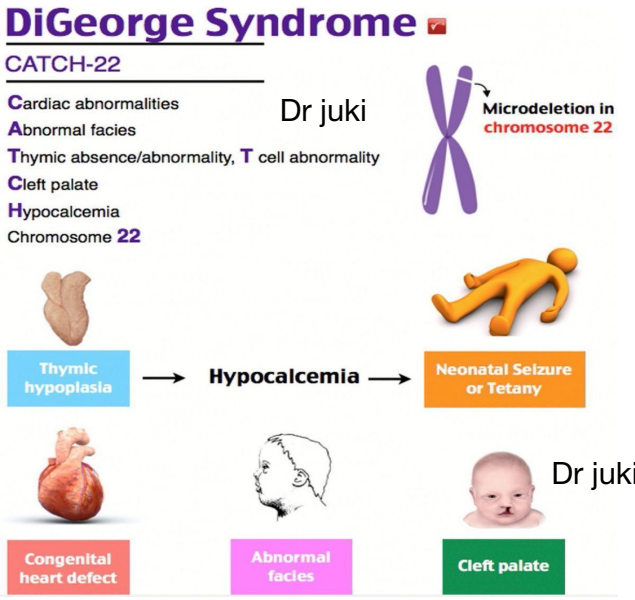
1. IV fluids (hydration) + furosemide
2. IV bisphosphonate, e.g, pamidronate.
3. Steroids are effective in hyperCa of sarcoidosis, vit D intoxication, malignancy & SC fat necrosis

Rx of 1ry Hyperparathyroidism (adenoma) → Surgery

Other causes of hyper Ca:

1. vit D toxicity
2. excess ca intake
3. drugs (thiazide diuretic –hypervitaminosis A)
4. syndromes (William syndrome)
5. ectopic (Tb –sarcoidosis – Addison)
6. prolonged immobilization.

DiGeorge Syndrome



. Facial anomalies associated with DiGeorge syndrome. Note the wide-set eyes, low-set ears, and shortened structure of the upper lip

تنبيه:
هذا العمل تم بعد جهد كبير من قبل الدكتور الاخصائي: عبدالرحمن الجعكي، ويعتبر ثمرة سنين من خبرة ممارسة طب الأطفال، كل ذلك كان ممزوج بحنكة العطاء الاكاديمي للمادة.
لهذا لا يجوز نسخ أو اقتباس أي جزء من المحتوى الآ بعد التواصل مع الناشر: الدكتور عبد الرحمن الجعكي.

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